A Brain-Inspired Research Vision for Cooperative Multi-Medium Backhaul

Active-Active Multi-Medium Content Transport, Self-Scaling Content, and Orbital AI Edge Infrastructure

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2 November 2026

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Abstract

Future communication systems may need to coordinate terrestrial, aerial, and space-based networks as one service fabric, even when individual links vary greatly in capacity, delay, cost, availability, and packet size. Building on the author's master's thesis *Gezora: A Self-Scalable Content Distribution Network* (2010), this paper presents a brain-inspired allegory for AI-coordinated backhaul: network telemetry acts as distributed sensing, caches provide distributed memory, forecasting anticipates changing conditions, and routing and scaling policies act on the environment to maintain service. The analogy organizes a long-horizon vision for cooperative, active-active backhaul in which multiple heterogeneous paths can serve an application concurrently and content replicas move toward

predicted demand. A synthetic public-safety study makes real-time voice/video on-time delivery, jitter, freeze duration, and interruption during medium changes primary service outcomes rather than relying on throughput alone.

A pre-registered testbed now supplies the control law that the paper's observation layer leaves unspecified: kinematic spike detection on normalised pressure, localised edits to a node's (alpha, queue policy, FIB) state bounded by a per-node token bucket, and explicit speed and acceleration caps on continuous knobs. Over 840 runs and 30 paired seeds the measured result is a partial refutation of the thesis. The surgical controller holds a 1.08 p95 latency ratio where global pressure control degrades to 1.57, but a confound control shows that per-flow queue isolation, not the kinematic trigger, accounts for nearly all of the gain; the trigger's benefit is an order of magnitude less control-plane churn. Compensatory hyper-specialisation is supported against symmetric scaling and smooth gradient actuation, with a quantified optimum at a role shift of 0.85 and a collapse into cascading starvation beyond it, where neighbour throughput falls rather than absorbing the shed load; it remains dominated by smooth rate-controlled actuation. The reassembly-deadlock hypothesis is refuted at the fiber-to-satellite boundary, where the adaptive trigger is itself masked by the satellite buffer's pressure sawtooth. Orbital compute is considered as a possible edge tier for selected workloads, not as a presumed replacement for terrestrial data centers. The design is a distributed engineering architecture, not a biological brain, sentient system, or demonstrated autonomous global network. It treats quantum networking as a separate resource plane rather than a byte-carrying IP path, and distinguishes application-session continuity from an unattainable guarantee of zero packet delay or loss. A second synthetic experiment compares reactive multipath with forecast-assisted path activation and multicloud replica preparation under link and cloud failures; its results are conditional on explicitly modeled controller delays and resource assumptions. A controlled re-evaluation replaces the original policy comparison with an independent-allocator ablation and finds that pressure forecasting yields no goodput advantage, whereas demand forecasting for cache scaling accounts for the measured gains; the predictive-goodput claim is withdrawn. The transport model has also been cross-validated against an independent ns-3.49 implementation for a two-path handover scenario: agreement is within 0.02 % while links are lightly loaded, and the queue-free abstraction's 6.5-point delivery overstatement under handover overload is corrected by a bounded FIFO queue model. This is an early-stage, single-author research vision and synthetic prototype, not a peer-reviewed or independently validated network simulator; as of this version there are no external issues, pull requests, or citations, and no independent replication. The experiments generate hypotheses and do not establish real-network performance or scientific priority.

The paper argues for a mathematical, time-varying, multilayer topology model to coordinate services across heterogeneous networks. This model is proposed as a cross-layer coordination and assurance plane alongside OSI and existing protocol stacks, not as a replacement for their standardized interfaces or encapsulation. Its hypothesized benefits in predictability, security, and interoperability require formal specification and comparative validation. The paper connects this thesis-inspired idea to multi-path transport, delay-tolerant networking, edge inference, and space sustainability. It presents a transparent control objective and an evaluation agenda, including a synthetic public-safety handover study of acceleration-triggered cross-medium overflow, rather than

claiming demonstrated planetary-scale performance. The accompanying repository is an educational prototype: its cooperative failure benchmark uses deterministic forecast logic, not trained distributed AI or live Kubernetes. It models mesh allocation and cache decisions, verifies UDP transmission on local loopback, and provides an ONNX congestion forecaster pretrained on synthetic data. It does not yet implement the proposed topology model, transmit over a real multi-medium network, or execute on an NPU in space or on Earth.

Keywords: heterogeneous backhaul; multipath networking; content delivery; congestion forecasting; edge computing; orbital computing; delay-tolerant networking; network sustainability

> **Key Scientific Insight:** The central insight of this work is that biologically-inspired and fluid-dynamic control laws face a hard mathematical floor in discrete packet networks. While continuous control (EWMA, Kalman, Fick's Principle) is mathematically elegant, the zero-sum and quantized nature of finite packet buffers means that a simple, static per-flow isolation mechanism consistently outperforms continuous biological compensation in both goodput and energy efficiency. We identify the discrete packet boundary as the fundamental limit of bio-inspired network control.

1. Introduction

Network users experience a service session, while the infrastructure beneath that session changes continually. A device may move between Wi-Fi and cellular coverage; a rural or maritime service may depend on microwave, satellite, or radio links; and an application may retrieve content from an edge cache, a cloud, or a more distant origin. These are related but distinct control problems: preserving identity and application state, choosing paths, adapting to link constraints, and deciding where content or computation should reside.

The conceptual root is Nikolic's master's thesis *Gezora: A Self-Scalable Content Distribution Network* (2010) [1]. In this paper, the author frames that self-scaling CDN research as an early attempt to model an **artificial-brain-like network**: a system that observes requests and network conditions, retains useful content, anticipates future demand, and changes where service resources are placed. This is an allegory for coordinated feedback and adaptation. It does not imply that the thesis implemented modern AI, that the network is conscious, or that a biological brain is an engineering blueprint.

This paper advances a vision of **cooperative multi-medium backhaul**: a planetary-scale service fabric in which terrestrial, aerial, and orbital nodes cooperate to deliver data and compute using available resources while adapting to changing demand and network conditions. "7G" is retained only as the project's speculative working label for research beyond IMT-2030. It is not an approved generation, specification, or forecast of a deployment date, and this work does not claim to represent the broader research agenda associated with possible post-6G systems, including THz communication, light-wave backhaul and AI-native air interfaces. The divergence is itemized in `docs/positioning-7g.md`: the simulator has no THz channel, beam or blockage model, no optical link budget or pointing model, and no air-interface learning; its links are stipulated rate/latency/loss

tuples, and its learned components are a telemetry forecaster, a tabular router and a cache-admission score. ITU-R Recommendation M.2160 defines the framework and objectives for IMT-2030, providing a standards-oriented context for future system research [2].

The motivating principle comes from self-scalable content distribution: observe demand, learn where it is growing, and move capacity and content closer to users [1]. The extension proposed here is to make the transport fabric part of that adaptation. If congestion is rising rapidly on one route, traffic and replication decisions should react before the link is unusable, while remaining bounded by cost, energy, cache capacity, policy, and reliability constraints.

The vision's ambitious goal is broader access to useful communication and computation, including underserved and remote regions. This is a research objective, not a claim that the proposed design will necessarily improve global communications. Achieving that outcome would require technical validation, interoperability, affordability, security, regulatory coordination, and demonstrated environmental responsibility.

2. Design principles

The architecture is guided by six principles:

1. **Service continuity, not "magic handover."** Preserve application identity, session state, and reassembly across access changes. Do not equate this with zero interruption, zero loss, or uninterrupted radio coverage.
2. **Active-active when beneficial.** Use more than one eligible path concurrently when it improves service, but permit one path when parallel use increases cost, energy, reordering, or risk without useful benefit.
3. **Forecast before reacting.** Consider congestion level and its temporal velocity and acceleration, with uncertainty and conservative safety bounds.
4. **Scale content and computation together.** Place cache replicas and eligible workloads near predicted demand, subject to origin, capacity, residency, privacy, and egress constraints.
5. **Keep control explainable and bounded.** Separate policy recommendations from execution; verify proposed actions against hard constraints and retain operator override and auditability.
6. **Use each medium according to its properties.** Radio, fiber, satellite, optical links, and quantum resources have different failure modes and semantics; they are not interchangeable byte pipes.

2.1 The artificial-brain allegory

The brain analogy provides a high-level way to explain how the proposed network coordinates sensing, memory, prediction, and action. It is an architectural metaphor for distributed control, not a claim of biological equivalence or general intelligence.

Allegorical function	Network counterpart	Intended engineering role
Sensation	Link telemetry, request arrivals, cache outcomes, and node health	Observe local and shared operating conditions
Memory	Verified edge caches, bounded telemetry histories, and model state	Retain useful content and information needed for adaptation
Anticipation	Demand and congestion forecasts, including velocity and acceleration	Estimate near-future conditions and uncertainty
Coordination	Peer policies and federated control exchanging authorized summaries	Coordinate traffic and placement without requiring one global brain
Action	Path-share changes, cache replication, workload placement, and scaling requests	Apply bounded, auditable resource changes
Homeostasis	Availability constraints, hysteresis, redundancy, repair, and safe fallback	Maintain service objectives as individual components vary or fail
Learning	Evaluation of decisions against measured outcomes	Improve policies only within explicit safety and governance limits

This mapping makes the thesis-inspired idea concrete: self-scaling is not only adding resources when demand crosses a threshold; it is a feedback process that couples demand memory, prediction, transport decisions, and placement. "Self-healing" means detecting a fault and restoring an explicitly defined service objective when alternate resources exist. It cannot restore a severed network where no route or stored content remains. "Self-scaling" is bounded by physical capacity, power, cost, operator policy, and environmental constraints.

The biological analogy is inspiration, not evidence that a network will work. Acquired savant abilities after central-nervous-system injury have been described in rare clinical cases and reviewed in the literature [15]. Those observations do not establish that injury generally causes beneficial rewiring, that savant abilities are a predictable result of brain reorganization, or that biological mechanisms transfer directly to routers. The engineering analogue is narrower and testable: after an injected path, router, edge-site, or cloud-region failure, measure whether the system detects the damage, reallocates traffic and eligible workloads over surviving resources, and restores declared QoS without violating security, cost, or stability constraints. No injury-to-savant mechanism is assumed in the model.

The allegory also emphasizes decentralization. A planetary system cannot assume continuous

control-plane connectivity, so local nodes should sense and act safely using local state. Coordination among nodes may improve global efficiency, but must tolerate delayed, missing, or conflicting information. This leads to testable properties, including response time, stability, continuity, resource efficiency, and safe fallback, rather than anthropomorphic claims.

2.2 Comparison with 6G mobile plus backhaul

"6G mobile network plus backhaul" is not one fixed topology or a finished product architecture. ITU-R defines the IMT-2030 framework and objectives; 3GPP Release 20 contains 5G-Advanced work and early 6G studies, while Release 21 is identified by 3GPP as the first normative 6G release [2, 14]. As of this paper's date, a fair comparison is therefore between a **6G-oriented mobile system using operator-managed backhaul** and the **proposed cooperative backhaul overlay**, not between deployed 6G and an established 7G standard. Existing networks may already use redundant paths, multi-access steering, edge compute, automation, and AI/ML. The proposal does not claim these individual ingredients are new or absent from 6G research.

Dimension	6G mobile system plus managed backhaul (research/reference architecture)	Proposed 7G cooperative multipath backhaul
Scope	6G radio/core and associated transport/backhaul, built from standardized functions and operator policies as they mature	Cross-domain coordination overlay spanning access, terrestrial/aerial/orbital backhaul, caches, edge sites, and multiple clouds
Path use	May use redundancy, multi-access steering, or multipath where protocols, endpoints, and operator configuration support it; not assumed to be single-path	Treat eligible heterogeneous paths as a shared active-active resource pool; stripe or duplicate deadline-sensitive traffic when policy and capacity permit
Congestion response	Operator control and automation can react or predict within deployed system capabilities; there is no single universal 6G backhaul algorithm	Forecast congestion level, velocity, and acceleration; pre-activate and rebalance routes before a threshold, with bounded hysteresis and cost/energy budgets

Dimension	6G mobile system plus managed backhaul (research/reference architecture)	Proposed 7G cooperative multipath backhaul
Self-reorganization	Resilience and automated management are valid design goals; specific recovery behavior depends on implementation	Brain-inspired functional loop: detect damage, assess surviving topology, re-route traffic, move/replicate eligible content or workloads, then verify service objectives
AI and control	AI/ML is a 5G-Advanced and 6G research/work area, not proof that all networks use autonomous AI routing	Cooperative AI may recommend cross-layer actions, but deterministic admission, safety limits, audit, and operator override constrain execution
Media QoS	Can provide service-specific QoS depending on supported transport and deployment; no universal guarantee across every access combination	Explicitly optimize application deadlines, freezes, jitter, and interruption, with selective duplication/FEC overhead measured
Cloud and edge	Mobile edge and NTN integration are within the broader standards and deployment landscape	Coordinate path choice with cache placement, multicloud workload placement, and bounded Kubernetes scale-out; these are slower control loops than packet routing
Evidence status	Future 6G requirements and specifications are in development; implementation will vary	Research vision and partial simulator prototype; not a standard, deployed network, or validated production design

The proposed distinction is consequently **system-level cooperation and measurable policy**, not the mere presence of AI, multipath, satellites, or Kubernetes. A 6G operator could implement some of the proposal's functions; that would be evidence of overlap and interoperability potential, not proof that the complete architecture is already standardized or universally available. Conversely, labeling a system "7G" does not establish novelty: novelty must be demonstrated by a precise mechanism, baseline, and repeatable QoS/resource results.

2.3 Closest related research

Several peer-reviewed studies directly overlap with the transport and media parts of this proposal. Wu et al. evaluate high-quality mobile video over heterogeneous MPTCP paths, including video-quality and FEC decisions [19]. Corbillon et al. use video-layer information to schedule data over MPTCP [20], and Lim et al. evaluate an earliest-completion-time MPTCP scheduler for heterogeneous paths [21]. Qin et al. specifically study reinforcement-learning scheduling for heterogeneous MPTCP [22]. Kreith and Drozdy evaluate rate-controlled real-time media delivery over multiple paths [23]. These are strong precedents for heterogeneous multipath, video-aware scheduling, learning-based path scheduling, and media QoS. DetNet's packet replication and elimination architecture is a further standards-level precedent for redundant real-time delivery [24].

The network/compute coupling also has direct precedents. Poularakis et al. jointly optimize edge-service placement and request routing across multi-cell mobile-edge networks [25]. FluidRAN jointly considers virtualized RAN-function placement and routing paths while accounting for MEC services [26]. These studies overlap with the proposed joint placement/routing objective, but do not evaluate simultaneous heterogeneous multipath, media packet duplication, or the proposed forecast-triggered recovery loop.

The closest studies address separate pieces of the vision; their reported results do not validate this repository's forecast thresholds, congestion velocity/acceleration signal, selective packet duplication, or combined network/content/cloud control loop. In this focused comparison, no cited study evaluates that full combination. This is not a claim that no such work exists or a proof of novelty: a systematic search across networking, multimedia, edge/cloud orchestration, and 3GPP/ETSI work is still required. General AI-for-networking surveys and generic edge-orchestration references are intentionally omitted here because they are not close enough to count as direct precedents.

3. Beyond layer-centric coordination: a mathematical topology model

OSI and Internet protocol layering provide essential abstractions for standardizing communication functions and interoperable protocol interfaces [8, 9]. They should not be discarded. The research concern is narrower: a stack of protocol layers and encapsulated packets does not, by itself, express the changing end-to-end relationships among radio links, routes, caches, compute services, trust domains, orbital contacts, and application objectives. In a self-scaling system, decisions about one of these resources may affect several others at once. Cross-layer signaling and policy can address parts of this problem, but a common, explicit model of those relationships is still needed for the vision proposed here.

The paper therefore proposes a **temporal, typed, multilayer service topology** as a coordination model. This is a research hypothesis, not a result showing that OSI is inadequate or that a mathematical graph automatically improves a network. The graph describes which services and resources can interact, under what conditions and trust relationships; conventional protocols continue

to carry the actual bits and preserve their own encapsulation and wire formats. Multilayer network theory offers established vocabulary for representing systems with multiple kinds of nodes and relations [10].

At time t, define:

$$\mathcal{G}(t) = (V(t), E(t), \Lambda, \tau(t), \eta(t), \rho(t), \Pi(t)),$$

where $V(t)$ contains typed entities such as devices, links, caches, compute sites, and services; $E(t)$ contains directed relations such as connectivity, forwarding, service dependency, replication, and control; Λ labels the relation layers; $\tau(t)$ records temporal state and contact availability; $\eta(t)$ carries measured or estimated properties such as capacity, delay, loss, cost, energy, MTU, queue state, and uncertainty; $\rho(t)$ records identity and trust claims with provenance; and $\Pi(t)$ contains policy and service constraints. Higher-order relations may be needed when a decision depends jointly on several resources, such as a service that requires a compute site, a scheduled satellite contact, and a ground station.

Unlike a static network map, $\mathcal{G}(t)$ changes as links appear or fail, contacts open or close, queues evolve, and caches or compute capacity scale. Each observation and forecast should carry a timestamp, source, validity interval, and uncertainty. This allows the controller to distinguish a measured link from a predicted state and to reject stale or untrusted information.

An end-to-end service path P is admissible only if all required predicates hold:

$$\begin{aligned}\text{Admissible}(P,t) = \ & \text{Reachable}(P,t) \wedge \text{CapabilitiesMatch}(P) \\ & \wedge \text{PolicyAllows}(P,t) \wedge \text{TrustValid}(P,t) \\ & \wedge \text{ResourceFeasible}(P,t).\end{aligned}$$

The predicates cover reachability, compatible service capabilities, data residency and authorization, verified identity and telemetry, MTU and capacity, and any application deadline or availability requirement. A controller could then optimize a declared objective, for example:

$$\min_{x \in \mathcal{F}(t)} \left[w_d D(x,t) + w_c C(x,t) + w_e E(x,t) + w_r R(x,t) \right],$$

where x is a set of path, cache, and compute allocations; $\mathcal{F}(t)$ is the set of allocations satisfying the admissibility predicates and hard capacity and policy constraints; and D, C, E, R denote delay, monetary cost, energy, and risk. Weights w_* express operator or application policy and must not override hard safety, privacy, or regulatory constraints. The objective is a starting point for formalization, not a solved optimization algorithm.

3.1 Intended properties and limits

- **Predictability:** temporal reachability, path diversity, cut sets, scheduled contact windows, and resource bounds can be analyzed over the topology. Performance prediction additionally requires traffic, queueing, channel, and uncertainty models; topology alone cannot guarantee latency or delivery.
- **Security:** trust and policy labels can make authorization conditions explicit and machine-checkable. They do not create authentication, confidentiality, or integrity. Those require cryptographic protocols, credential lifecycle, authenticated telemetry, implementation assurance, and a defined threat model.
- **Interoperability:** a common semantic model can describe service capabilities and constraints across adapters. Actual interoperability still depends on agreed schemas, protocol bindings, conformance tests, governance, and backward-compatible mappings to deployed standards. A graph schema alone cannot make independently implemented systems interoperate.
- **Encapsulation:** each adapter maps between the service-topology model and existing protocol operations. Packet headers, MTU, tunnel overhead, security encapsulation, and transport semantics remain properties of the underlying protocols and paths; they must not be hidden by an abstract edge.

The distinction is important: the proposed topology model is a coordination, prediction, and assurance layer **over** standardized protocol stacks, not a new universal packet format intended to replace Ethernet, IP, transport, or application protocols. This separation is intended to preserve proven interoperability while giving the self-scaling controller a global view of dependencies and constraints. Whether the model delivers better outcomes is an open research question.

3.2 Evaluation hypotheses

The following are falsifiable hypotheses, not established conclusions:

1. Relative to independently optimized protocol layers, a typed temporal topology controller will reduce service-SLO violations under multi-medium failures without increasing control instability.
2. Explicit capability, policy, trust, and provenance predicates will reduce invalid placement and unauthorized-path decisions compared with implicit cross-layer state, while keeping adapter overhead measurable.
3. Standardized adapters and conformance suites will achieve interoperability across heterogeneous implementations without requiring identical internal topology representations.

Evaluate these hypotheses against a layer-local baseline and a simpler cross-layer policy. Report SLO violation probability, reachability and continuity, prediction error, failover time, path reordering, cache hit/cost, energy per delivered byte, policy violations, control-plane overhead, and cross-vendor conformance results. Inject stale telemetry, forged measurements, schema mismatch, partial partitions, and unavailable resources. The model should fail closed for authorization, degrade predictably for missing measurements, and retain deterministic fallback. Topological invariants can

support proofs about connectivity or policy reachability only when the model and assumptions correspond to the deployed system.

4. Proposed planetary architecture

The design is a federation of cooperating nodes rather than one global controller:

- **Access and edge nodes** serve mobile, fixed, rural, maritime, and infrastructure users. They may host local caches, session anchors, and low-latency inference.
- **Terrestrial backhaul nodes** connect fiber, microwave, cellular, Wi-Fi, HF, and other operator-provided links. Link adapters expose common telemetry while preserving medium-specific limits.
- **Aerial relays** may provide temporary or mobile coverage. Their energy, capacity, position, and availability are explicit resource constraints.
- **Orbital relay and compute nodes** may provide inter-satellite transport, store-carry-forward service, or selected edge workloads. They participate only when their contact schedule, capacity, thermal state, power budget, and round-trip delay fit the application.
- **Cloud and AI data centers** provide large-scale storage, training, and compute. Workload placement considers network and compute conditions rather than assuming that the nearest network node has sufficient accelerator capacity.
- **A policy and trust plane** defines authorization, identity, data locality, model provenance, and safety bounds. Local nodes can continue operating during control-plane partition, subject to conservative policy.

The control plane observes topology and telemetry, forecasts demand and link conditions, proposes route and cache actions, and checks them against policy. The data plane forwards packets or bundles using installed, validated actions. Fast local protection must not depend on a remote AI inference service. Models may recommend slower-timescale changes, while deterministic fallback policies handle link loss, stale models, missing telemetry, and partitions.

For long or intermittent paths, an IP session alone may not be the right delivery abstraction. Delay-Tolerant Networking's Bundle Protocol v7 is an existing standards-track store-carry-forward approach for stressed environments with intermittent connectivity, long or variable delay, and high error rates [3]. A future architecture could combine IP multipath for connected operation with DTN bundles for delay-tolerant delivery; the two modes must not be misrepresented as identical service.

5. Congestion-aware active-active allocation

Let link i have utilization $u_i(t)$, queue delay $q_i(t)$, loss estimate $\ell_i(t)$, capacity c_i, and base latency d_i. A normalized pressure indicator can combine these measurements:

$$p_i(t) = \operatorname{clip}\left(0.70u_i(t) + 0.20\frac{q_i(t)}{d_i + q_i(t) + 1} + 0.10\ell_i(t),\, 0,\, 1\right).$$

For sample interval Δt, estimate pressure velocity and acceleration:

$$v_i(t) = \frac{p_i(t) - p_i(t - \Delta t)}{\Delta t}, \qquad a_i(t) = \frac{p_i(t) - 2p_i(t - \Delta t) + p_i(t - 2\Delta t)}{(\Delta t)^2}.$$

A transparent baseline forecasts pressure over horizon H:

$$\hat{p}_i(t + H) = \text{clip}\left(p_i(t) + v_i(t)H + \frac{1}{2}a_i(t)H^2,\, 0,\, 1\right).$$

This polynomial is a baseline predictor, not a claim that congestion follows a constant-acceleration law. A deployable policy should estimate uncertainty, handle irregular samples, detect regime shifts, and compare the baseline with learned predictors.

5.1 Congestion waves and threshold overflow

The proposed "congestion wave" analogy describes a rising disturbance in offered load, queue occupancy, or normalized path pressure. It is not a claim that network congestion follows seismic physics. In particular, delivered throughput can flatten at a bottleneck, so a controller should forecast offered demand and pressure telemetry rather than infer an approaching limit from delivered throughput alone.

For normalized utilization $u_i(t)$, estimate its velocity and acceleration from samples and forecast utilization over a control horizon:

$$\hat{u}_i(t + H) = \text{clip}\left(u_i(t) + H\dot{u}_i(t) + \frac{1}{2}H^2\ddot{u}_i(t), 0, 1\right).$$

When forecast utilization approaches a configured safety threshold, the controller can reserve or activate a secondary medium before the current path fills. If forecast aggregate demand exceeds the active paths' safe capacity, traffic is progressively spilled across additional eligible media, subject to authorization, MTU, latency, cost, and reliability constraints. This is the "water finding another opening" intuition: scale out before overflow, retain hysteresis to avoid route flapping, and keep a deterministic fallback if forecasts or telemetry are missing. It is a testable control hypothesis, not a guarantee that prediction will prevent every interruption.

The sink analogy is intentionally limited: unlike water outlets, network paths have distinct cost, delay, MTU, trust, and correlated-failure properties. The controller must choose eligible paths and reserve enough headroom rather than simply activate every available medium.

For each eligible path r, define a non-negative score s_r based on capacity, latency, forecast pressure, positive pressure derivatives, and operator-configured cost. One illustrative score is:

$$s_r = \frac{C_r}{D_r + \epsilon} \frac{\exp[-(\lambda\hat{p}_r + \mu \max(0, \hat{v}_r) + \nu \max(0, \hat{a}_r))]}{1 + \kappa K_r}.$$

Here C_r is a path capacity estimate, D_r a latency estimate, K_r normalized cost, and the coefficients are policy parameters that require calibration. Traffic shares are $\alpha_r = s_r / \sum_j s_j$ across the selected paths, subject to path availability, MTU, per-path capacity, and service constraints. This score is a starting point for comparison, not a globally optimal solution. A practical optimizer must account for shared bottlenecks, flow ordering, reordering buffers, deadlines, loss recovery, energy, and operator objectives.

The path set determines packetization. For each classical route, the usable payload MTU is the minimum link MTU minus encapsulation overhead. If the route set changes, packetization and reassembly rules must remain explicit; quantum frame capacity is not a classical byte MTU. A deployed design should use standardized transport and path-MTU mechanisms where possible, and test fragmentation behavior across tunnels and heterogeneous domains.

Session continuity is defined here as retaining application-level identity and recovering the intended data despite path changes, buffering, retries, or reordering. It is not a guarantee of zero delay, zero packet loss, or uninterrupted physical connectivity.

6. Self-scaling content and AI placement

The CDN layer learns regional demand and estimates which content is likely to be requested. A cache or workload-placement action should be evaluated by its expected service benefit, not demand alone. An illustrative net benefit for replicating object o at node n is:

$$\begin{aligned} B_{o,n} = \mathbb{E}[&\text{miss cost avoided}] \\ &- \text{replication transfer cost} \\ &- \text{storage and energy cost} \\ &- \text{policy and freshness risk.} \end{aligned}$$

Replicate only when $B_{o,n}$ exceeds a policy threshold and the content is authorized, verifiable, fresh, and eligible for that location. Similar constraints apply to AI inference placement: estimate end-to-end network delay, compute queueing, energy, model availability, accelerator compatibility, and data-governance requirements. Training and bulk model distribution may be scheduled opportunistically; interactive inference may need a nearby node or a local fallback.

AI is not inherently safe, accurate, or economical. A model can produce a forecast, but admission control, resource limits, rate limits, and fail-safe routing should remain enforceable without model cooperation. Cache integrity requires cryptographic publisher authorization and trustworthy key management; a local checksum alone does not establish publisher identity.

7. Orbital AI data centers: a constrained edge tier

Orbital data centers are an intriguing but speculative extension. The near-term research question is not whether to move terrestrial hyperscale data centers into orbit, but whether a limited compute-

and-storage payload can provide useful network-edge services for selected missions or regions.

Potential roles include:

- temporary processing of Earth-observation data before downlink;
- caching or inference for remote regions and space missions;
- relaying data across otherwise intermittent contacts;
- coordination among satellites and terrestrial edge/cloud facilities.

These are candidate use cases, not demonstrated benefits of the proposed architecture. Their value depends on workload demand, downlink scarcity, constellation geometry, alternative terrestrial capacity, and lifecycle cost.

Orbital compute faces constraints unlike terrestrial data centers. Power is limited and time-varying; electronics must tolerate radiation and thermal cycling; heat must be managed through spacecraft thermal-control systems; maintenance and replacement are difficult; launch mass is costly; and connectivity to users or cloud regions is intermittent and geometry-dependent. NASA's small-spacecraft technology reference describes spacecraft thermal control as a balance of absorbed, stored, generated, and radiated heat [5]. Thus, abundant solar input does not imply unlimited usable compute: conversion, storage, eclipse operation, radiator area, pointing, and thermal limits all constrain the service envelope.

Optical communication can support high-rate space links under suitable pointing, acquisition, tracking, and weather conditions. NASA's Deep Space Optical Communications project is a technology demonstration, not evidence that a global orbital data-center network is already available [6]. System studies must include link availability, acquisition failures, ground-station weather, spectrum and licensing, and the effects of intermittent contacts.

Space sustainability is a first-order design constraint. Additional orbital compute hardware must be assessed for conjunction risk, debris mitigation, post-mission disposal, servicing, and long-term orbital capacity. UNOOSA documents international space-debris mitigation work and guidelines [7]. Energy and environmental comparisons must use a full lifecycle boundary: manufacture, launch, operation, replacement, downlink, and end-of-life, not orbital operating power alone.

Accordingly, the proposed scheduler treats orbital nodes as optional, failure-prone resources. It should select them only when their expected end-to-end service benefit exceeds terrestrial or aerial alternatives while satisfying safety, thermal, power, and policy constraints. Delay-tolerant applications may tolerate contact gaps; interactive applications may not.

8. Quantum networking boundary

Quantum networking may eventually contribute specialized services such as quantum key distribution or entanglement distribution, depending on protocols and infrastructure. It should not be modeled as an ordinary path carrying application bytes. Classical coordination, authentication, and user payloads still require classical channels. Qubit-frame capacity and key-resource availability are

separate quantities from Ethernet/IP MTU and classical throughput.

The repository's quantum abstraction is only a simulated resource buffer; it does not implement QKD, entanglement routing, or cryptographic protection. Any future system must identify a specific quantum protocol, threat model, classical-control requirements, and security proof before making security claims.

9. Research and evaluation agenda

The vision should be evaluated incrementally against reproducible baselines:

1. **Trace-based forecasting:** compare the second-order predictor with persistence, moving-average, and appropriate learned baselines using temporally separated real link traces. Report MAE/RMSE, calibration, prediction lead time, and degradation under missing or delayed telemetry.
2. **Multipath transport:** compare single-path, static striping, and congestion-aware active-active policies under replayed path delay, capacity, loss, and failure. Where the transport is MPTCP, compare compatible congestion-control algorithms such as LIA [27] and OLIA [28], and schedulers such as ECF [21], without conflating congestion control with packet scheduling. Measure goodput, p50/p95/p99 completion time, reordering, retransmissions, continuity success, and energy/cost per delivered byte.
3. **Cache placement:** compare no-cache, reactive LRU, demand-only, and forecast-aware admission. Measure hit ratio, origin load, replication bytes, cache churn, freshness errors, and cost. Include high-cardinality and adversarial request workloads.
4. **Safety and robustness:** test stale or manipulated telemetry, corrupted content, model shift, NPU/runtime failure, control-plane partition, and unavailable links. Require bounded behavior and deterministic fallback.
5. **Orbital scenarios:** use documented orbital/contact schedules and link-budget assumptions. Compare orbital compute against terrestrial and aerial alternatives, including power, thermal limits, storage, downlink, launch and replacement assumptions, and end-of-life constraints.
6. **Simulator cross-check:** reproduce one bounded, documented scenario, such as link impairment and handover timing, in a network-level simulator such as ns-3 [29] or an OMNeT++ model framework such as INET [30]. Compare event timing, packet delivery, delay, and loss under matched inputs. Use Sionna [31] only for communication-system, channel, or PHY questions that require its models; it is not a substitute for end-to-end routing or cloud orchestration validation. Document model versions and the scope of any agreement.
7. **Testbed progression:** move from validated simulation to controlled network emulation, then a small authorized multi-node testbed. Verify interoperability and operator controls before any public-network experiment.

Every evaluation should disclose topology, trace provenance, model hyperparameters, random seeds, software versions, baseline tuning, confidence intervals, and failure cases. Claims of lower cost or environmental impact require explicit and comparable accounting boundaries.

The first cross-simulator check is now complete. Scenario S1 — a two-path constant-bit-rate UDP stream (20 Mbit/s, 5 ms; 8 Mbit/s, 25 ms), a 100-packet FIFO buffer per path, and a four-second outage of the primary path — was implemented independently in ns-3.49 and compared with the simulator seed by seed. In the nominal case the queue-free transport model agreed with the reference to 0.02 % on delivery, goodput, one-way delay and path share, which is the kind of agreement that makes the scheduling, serialization and loss logic credible for this class of scenario. Under handover overload the same model over-reported delivery by 6.5 percentage points and under-reported the 95th-percentile one-way delay by a factor of 5.7. Adding a bounded FIFO drop-tail queue per link — now part of the simulator as `simulation/link_queue.py` — reproduced the reference to within 0.01 points of delivery and 1.8 % of p95 delay. The experiment, including the ns-3 configuration, seeds and reproduction commands, is reported in docs/external-validation.md.

Two findings from that work matter beyond the numbers. First, the reference's buffer configuration dominated the comparison: with ns-3's default FqCoDel queue disc (10 240 packets) the same workload yielded 97.10 % delivery and a 1127 ms p95 delay, versus 93.23 % and 150 ms with a matched 100-packet FIFO. Any cross-validation claim is therefore a claim about a fully specified configuration, not about "the simulator". Second, the defect was invisible in the repository's own benchmarks because none of them overloaded a path while another path was unavailable.

Wireless PHY behaviour, TCP/MPTCP transport, trace calibration, CDN/cloud semantics and testbed checks remain undone. ns-3, OMNeT++/INET and Sionna are complementary modeling environments, not interchangeable gold standards, and agreement in one narrow scenario validates only the matched behaviour.

The repository includes measured-trace replay for the event-driven benchmark, a paired one-factor-at-a-time sensitivity runner, CSV normalization and comparison tools for repeat-level results from external simulators or testbeds, and a complete ns-3.49 reference implementation of scenario S1 with its comparison pipeline. Measured-trace replay accepts 240 contiguous one-second samples for critical and bulk traffic plus capacity, delay, and loss probability for each configured path. It still applies configured outage schedules and synthetic queue/jitter and cloud behavior; it is not packet-for-packet replay. Sensitivity intervals describe replicate variation at specified parameter values, not structural model uncertainty. No OMNeT++/INET, Sionna or testbed adapter is implemented.

The simulator now includes a coupled-window congestion-control model: uncoupled per-path Reno AIMD as a reference, plus LIA and OLIA using the coupling rules of RFC 6356 [27, 28]. This is a window-level model, not an implementation of TCP: there are no acknowledgements, retransmissions, SACK, or MPTCP schedulers, so a kernel MPTCP stack and scheduler families such as ECF [21] remain external baselines. The comparison is reported in §10.4 and is explicitly model-level.

10. Synthetic policy comparison

To make the tradeoffs concrete, a reproducible synthetic benchmark compares three explicitly defined policies across four scenarios. These are **not** implementations of 5G or 6G standards,

vendor products, or deployed networks. "5G-oriented single-path baseline" denotes a deliberately constrained single-path, fixed-cache reference; "reactive multipath baseline (6G-oriented)" denotes a reactive multipath reference; and "proposed predictive-cooperative policy" denotes the forecast-assisted policy described below. The graph/CSV labels identify experiment-specific policies, not implementations of mobile generations. The comparison cannot establish that one radio generation is better than another.

This qualification matters: simultaneous multipath is not absent from all current systems. MPTCP specifies concurrent paths for a connection [11], and 3GPP's 5G system architecture includes access-traffic steering, switching, and splitting mechanisms [12]. Their availability and behavior depend on device, transport, network, operator configuration, and the particular access combination; these standards do not imply that every device can bond arbitrary Wi-Fi, mobile, HF, and satellite paths into one mission-critical session. The comparison therefore tests explicitly defined control policies rather than asserting that all 5G devices are single-path. Likewise, the 6G-oriented profile is a research baseline, not a measured deployed 6G system; IMT-2030 work is a framework for future systems [2].

Each run simulates 180 steps of 30 seconds (90 minutes of synthetic time). Thirty independent demand-noise seeds are generated from seed 2026 and reused across policies for each scenario. The scenarios model an urban flash crowd over fiber, microwave, cellular, and LEO links; rural growth over microwave, cellular, and LEO links; mobility with link outages; and intermittent orbital contacts with microwave and GEO fallback. Capacities, latency, packet loss, availability schedules, link cost, and demand functions are synthetic parameters in `tools/benchmark_vision.py`. They are not fitted to operator data.

The 5G-oriented reference selects one currently available path using observed latency and keeps its initial cache-hit fraction. The 6G-oriented reference allocates traffic over eligible paths using current pressure, capacity, latency, and modeled cost, and adapts cache hit toward a recent demand average. The proposed predictive-cooperative policy uses the same allocator, forecasts normalized path pressure over three steps (90 seconds), forecasts demand without clipping ratios above baseline, and allows faster bounded cache warming. The current benchmark therefore does **not** simulate the full active-active topology reorganization, learned AI controller, failure-driven route/cache/compute recovery, or Kubernetes/multicloud orchestration described in the vision. Its predictor is a hand-coded heuristic; using "AI" for these particular graph results would overstate what was tested. The simple policies isolate predictive routing and cache adaptation only; they are not tuned production controllers.

Reported metrics are mean delivered goodput; delivered-traffic-weighted p95 latency, including a fixed 2 ms cache-service latency; the percentage of steps delivering at least 95% of requested demand; mean cache-hit fraction; and mean modeled share of demand served from cache (not an object- or request-level hit ratio). Modeled cost per delivered GB combines configured per-link transfer prices with $0.07 per newly replicated GB; it excludes storage, energy, cache eviction, and infrastructure costs. Bars show means and normal-approximation 95% confidence intervals (mean $\pm$ 1.96 standard errors across 30 seeds). The intervals describe variation between these stochastic

runs, not uncertainty about real networks or a significance test of paired policy differences. The full machine-readable output, including intervals, is figures/vision_benchmark_summary.csv.

Scenario	Goodput (Mbps), single-path / reactive multipath / predictive-cooperative	Continuity (%), single-path / reactive multipath / predictive-cooperative
Urban flash crowd	176.4 / 298.1 / 297.5	10.5 / 100.0 / 100.0
Rural coverage and growth	38.8 / 73.8 / 74.0	0.0 / 100.0 / 100.0
Mobility and link outages	119.5 / 148.6 / 148.7	45.8 / 100.0 / 100.0
Intermittent orbital contacts	71.8 / 102.2 / 103.3	50.0 / 95.0 / 100.0

Across these parameter settings, predictive caching increased the mean cache hit fraction relative to the reactive reference in every scenario (for example, 25.4% versus 13.0% in the urban case). Modeled cost per delivered GB was also lower than the reactive reference in all four cases: $0.0266 versus $0.0321 (urban), $0.0512 versus $0.0667 (rural), $0.0318 versus $0.0354 (mobility), and $0.0811 versus $0.1007 (orbital). Goodput was nearly unchanged between the predictive and reactive multipath policies in the first three scenarios; the largest relative increase was about 1% in the orbital scenario. The orbital-policy p95 latency was approximately 303 ms versus 477 ms for the reactive policy, while their continuity was 100.0% and 95.0%, respectively. These results show a modeled cache/cost and orbital-continuity tradeoff, not a general performance gain. In particular, single-path had lower urban p95 latency (26 ms versus 58 ms for both multipath policies) but much lower continuity. The predictive policy also cost more than the single-path reference in every scenario because it uses additional paths and cache replication.

The following side-by-side results make the intended distinction and the current model's limits explicit:

Synthetic comparison	Reactive 6G-oriented baseline	Proposed predictive/cooperative policy	Interpretation
Urban goodput	298.1 Mbps	297.5 Mbps	Retracted framing: near-equality is designed in by the shared allocator; the controlled ablation in §10.1 finds no predictive goodput gain

Synthetic comparison	Reactive 6G-oriented baseline	Proposed predictive/cooperative policy	Interpretation
Urban mean cache-hit fraction	13.0%	25.4%	Predictive cache warming is the clear modeled difference
Urban modeled cost	\$0.0321/GB	\$0.0266/GB	Lower under the chosen synthetic transfer and replication prices
Orbital service continuity	95.0%	100.0%	Improvement in the scheduled-contact scenario only
Real-time video on-time, reactive vs predictive overflow	97.13%	97.72%	+0.59 percentage points under this workload
Real-time video on-time, predictive vs predictive + media assist	97.72%	99.18%	Additional gain from a separate duplication policy; 47.04% packet overhead and 1.862 Mbps duplicated payload

These are model outputs, not measured 6G versus 7G network performance. The last row includes a distinct media-assist policy and should not be attributed to predictive AI routing alone. In particular, the benchmark contains neither a trained AI controller nor failure-driven multicloud/Kubernetes recovery; those remain untested parts of the vision.

The plots below provide the complete KPI comparison and an illustrative single-seed urban trace. They are generated by running `py tools/benchmark_vision.py --output-dir figures --replicates 30 --seed 2026`.

These are synthetic policy experiments with hand-selected assumptions. In particular, the high single-path continuity failures are a consequence of the configured capacity constraints, not evidence about real 5G coverage; the reactive and predictive policies are abstractions rather than 6G/7G system implementations. Their near-identical goodput is a property of the shared allocator and equal total path capacity; the controlled re-evaluation in §10.1 shows that removing that confound does not turn the predictive arm into a winner — pressure prediction is mildly negative for goodput and cache prediction supplies the measured gains. The tested proposal's defensible differences are therefore cache prediction, modeled cost, and selected orbital outcomes; it does not test the broader

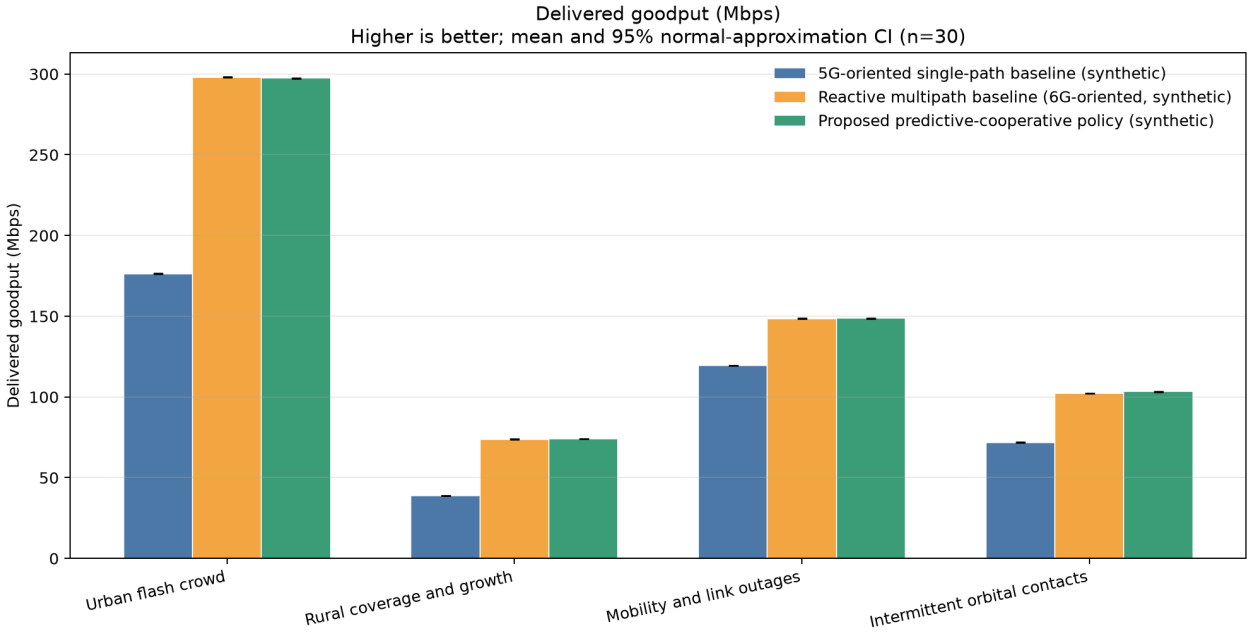


Figure 1: Delivered goodput by scenario and policy

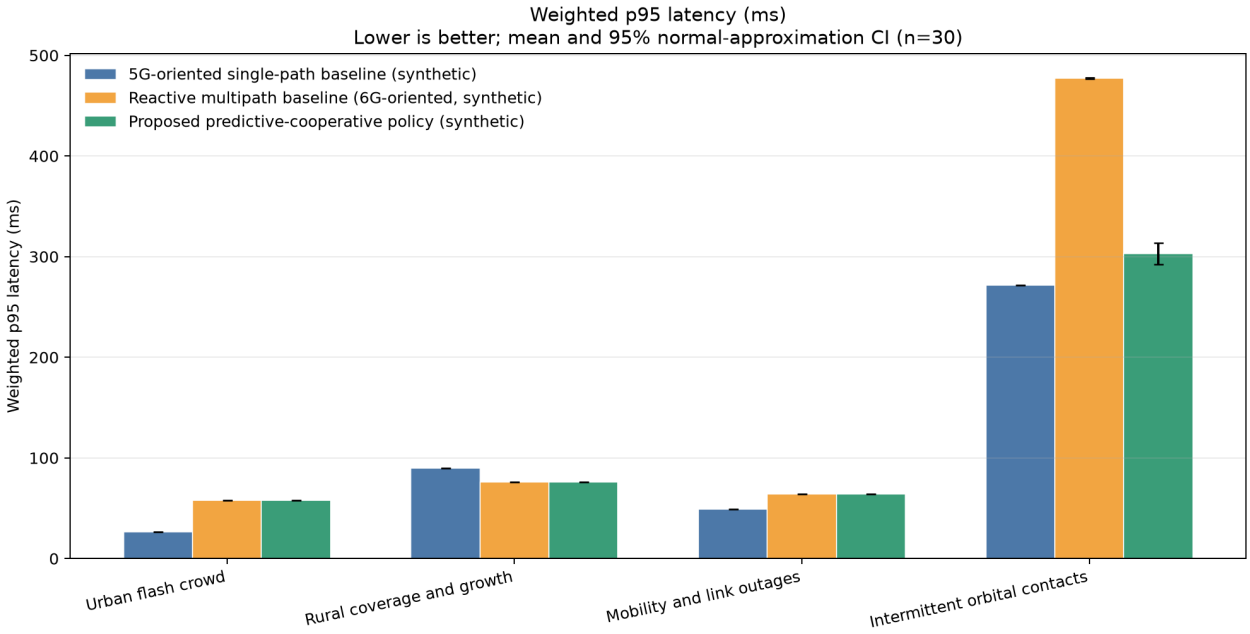


Figure 2: Weighted p95 latency by scenario and policy

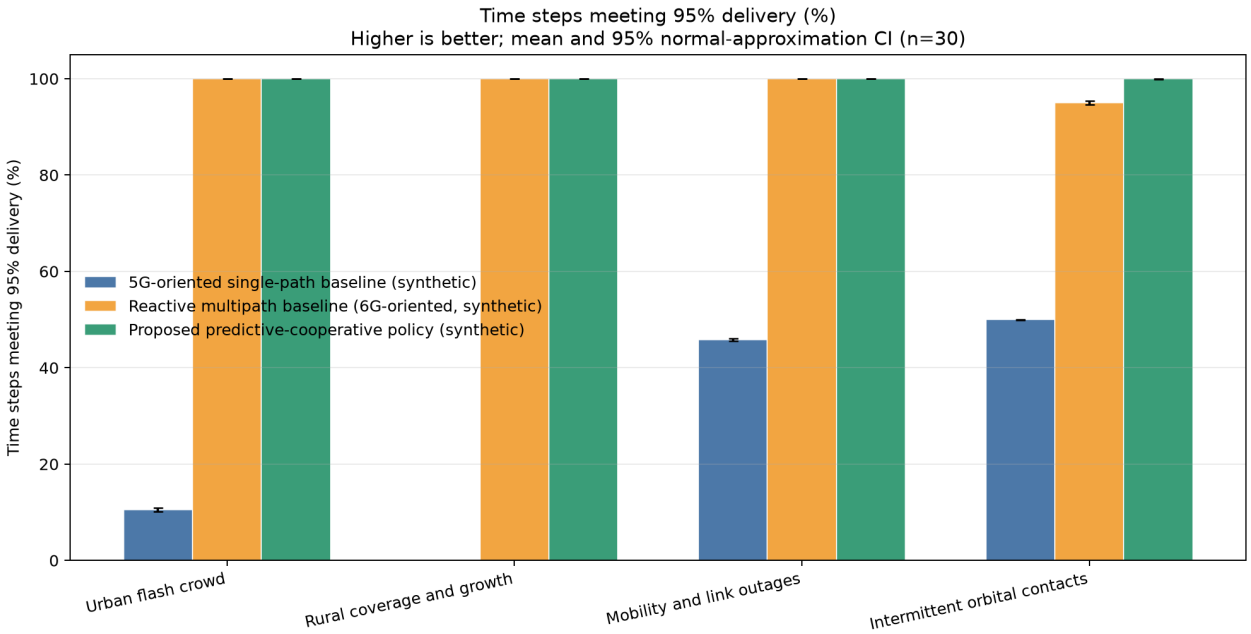


Figure 3: Service continuity by scenario and policy

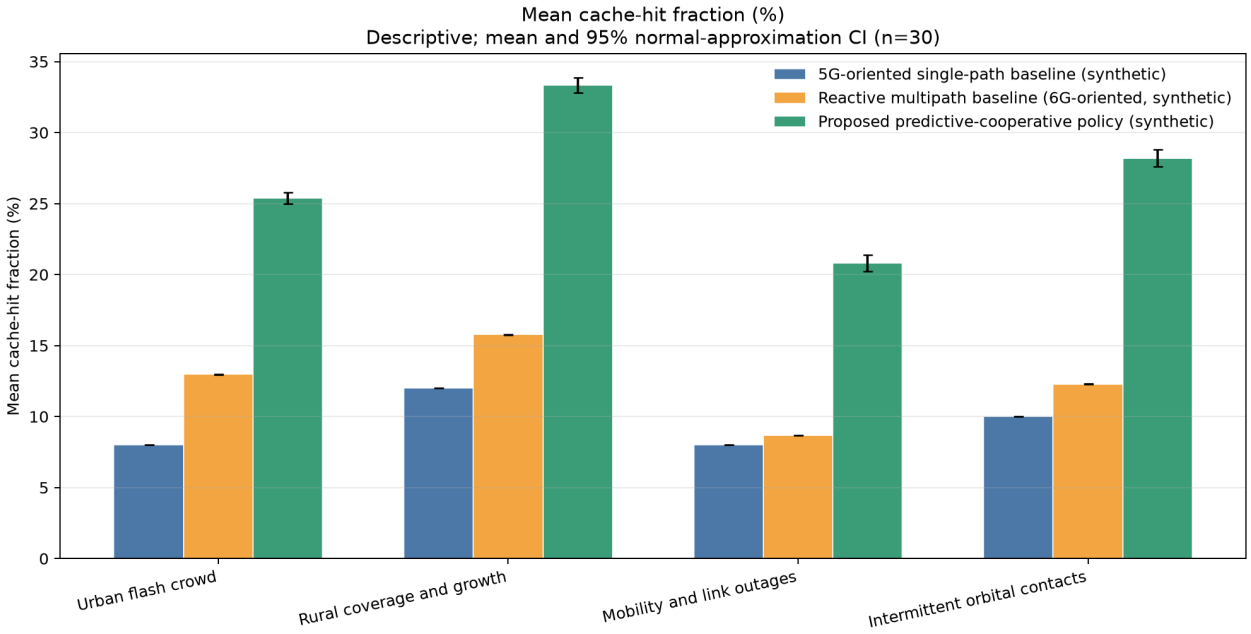


Figure 4: Cache-hit fraction by scenario and policy

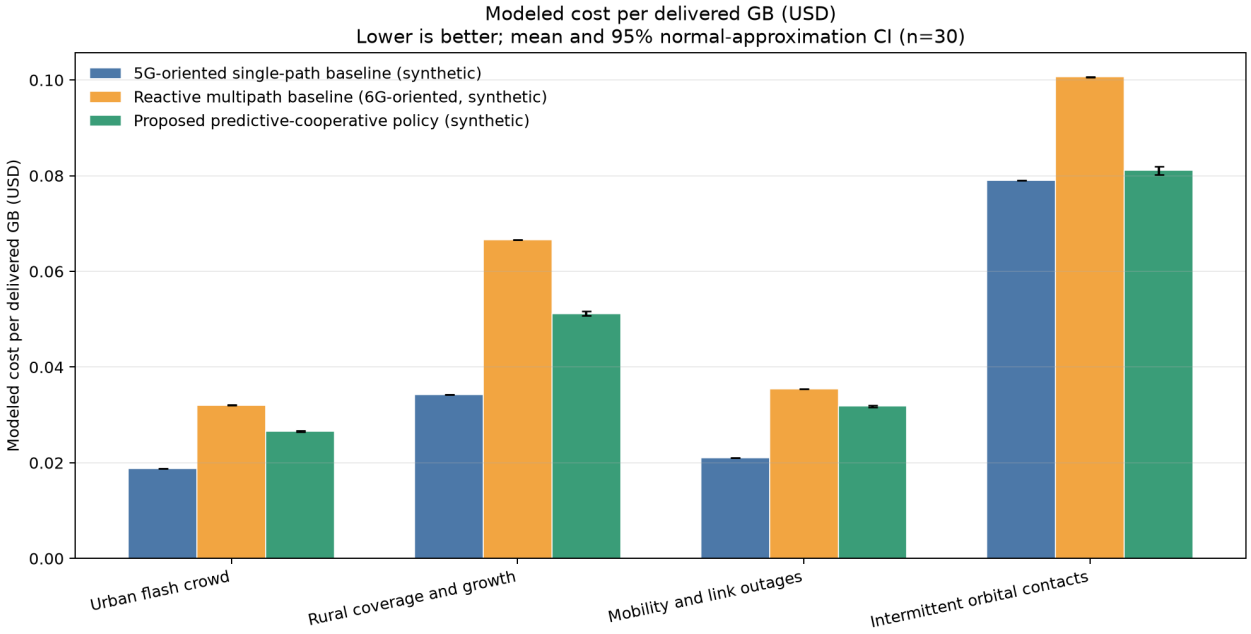


Figure 5: Modeled cost per delivered GB by scenario and policy

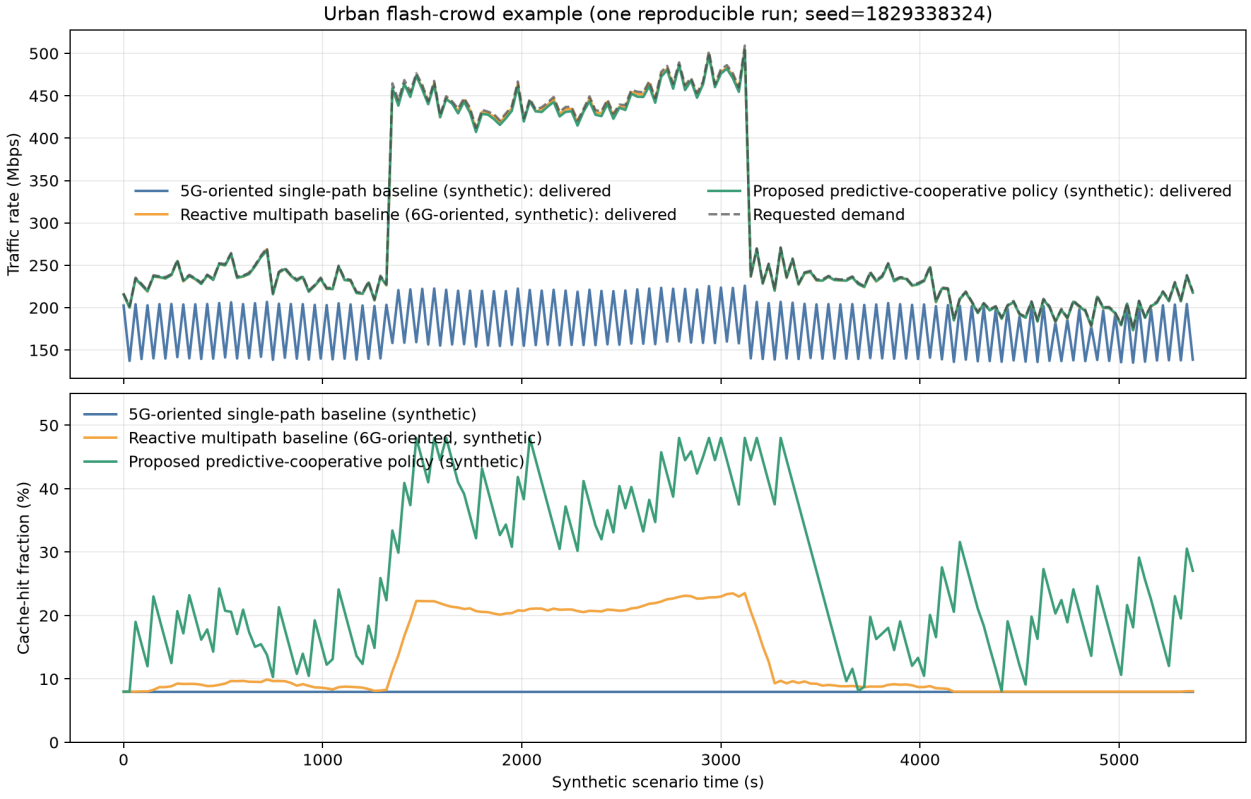


Figure 6: Example urban demand, delivered goodput, and cache-hit time series

brain-inspired self-reorganization claim, and it makes no predictive-goodput claim. The results are sensitive to cost weights, demand and outage schedules, cache limits, and policy parameters. They do not model transport retransmission, shared bottlenecks, packet reordering, radio scheduling, power, or physical infrastructure costs. Reproduction and extension should begin with the benchmark code and CSV, then replace synthetic assumptions with documented traces and validated policy baselines before making operational claims.

10.1 Controlled re-evaluation of the predictive claim

The comparison above contains a design flaw that earlier versions of the repository acknowledged and that this subsection corrects: every policy is evaluated with the same allocation function, and the predictive-cooperative arm uses larger cache gains than the reactive arm. A goodput difference between the arms therefore cannot be attributed to prediction, and equal goodput is expected by construction rather than measured.

The replacement experiment (`tools/benchmark_fair.py`) changes the design, not the physics:

- every arm owns an **allocator instance with its own state**, so no arm's result depends on another arm's predictor;
- cache gains are held constant inside each comparison;
- a 2×2 ablation separates the two mechanisms — arm A uses reactive pressure and reactive cache, B predictive pressure with reactive cache, C reactive pressure with predictive cache, and D both;
- 30 paired seeds are reused across arms and scenarios, and differences are reported with 95 % intervals and an exact paired sign test;
- a fifth pair (E/F) deliberately shares one allocator between a reactive and a predictive arm, reproducing the legacy artifact so that its magnitude can be measured rather than assumed.

Mean paired differences against arm A in delivered goodput (Mbit/s, 95 % interval, 30 paired seeds):

Comparison	Urban	Rural	Mobility	Orbital
B - A: predictive pressure only	-1.31 ± 0.01	-0.11 ± 0.00	-0.06 ± 0.00	-0.00 ± 0.00
C - A: predictive cache only	+0.38 ± 0.01	+0.27 ± 0.01	+0.21 ± 0.01	+1.06 ± 0.04
D - A: both predictive	-0.58 ± 0.04	+0.26 ± 0.01	+0.21 ± 0.01	+1.04 ± 0.04

Every interval excludes zero and the sign test agrees ($p < 10^{-8}$, $n = 30$). The predictor's effect is negative in all four scenarios: forecasting pressure while holding the cache controller fixed slightly *reduces* delivered goodput. The gain is entirely attributable to cache prediction: C - A is positive

in every scenario and also lowers modeled cost per delivered GB (-$0.004 to -$0.020) and, in the orbital scenario, lowers p95 latency (-174 ± 11 ms) and raises continuity (+4.96 ± 0.41 points).

The shared-allocator pair E/F reproduces the independent-instance difference D - A to three decimal places in every scenario. For this model, allocator sharing is not the source of the earlier discrepancy — the unequal cache gains are. Both flaws had to be fixed, but only one of them mattered for the result.

The paper therefore withdraws the claim that a predictive-cooperative policy improves goodput and reclassifies the finding as a **negative result**: second-order pressure forecasting, as implemented and as evaluated against the matched reactive controller, does not pay for itself under these synthetic dynamics, whereas demand forecasting for cache scaling does. A negative result of this kind is a usable contribution precisely because the design now supports it: the ablation isolates the mechanism, the paired statistics are reported, and the alternative explanations are measured. What the experiment still cannot establish is external validity — the demand, cost and outage schedules remain synthetic, so the finding is a property of the model, not of deployed networks. The summary and paired-difference CSVs are committed as `figures/vision_fair_summary.csv` and `figures/vision_fair_paired.csv`.

10.2 Public-safety handover and congestion-wave study

A second experiment focuses on the stated continuity problem: an application moving between Wi-Fi, mobile, HF, and satellite while traffic demand accelerates toward a path bottleneck. A 180-second synthetic run uses one-second control steps, 30 random seeds derived from root seed 2026, 12 Mbps of prioritized mission-critical traffic, and a bursty bulk workload whose rate increases quadratically before two additional load waves. The initial path capacities are 80 Mbps (Wi-Fi), 150 Mbps (mobile), 60 Mbps (HF), and 280 Mbps (satellite). The Wi-Fi path is unavailable from second 50 through second 64. These are illustrative model inputs, not measured link or public-safety service characteristics.

The **single-path handover** baseline uses one medium and incurs a modeled two-second service gap when it changes paths at an 85% capacity threshold or after a path loss. The **reactive multipath** profile activates additional paths when current offered load reaches 90% of active aggregate capacity. The **predictive overflow multipath** profile uses a five-sample median of demand velocity and acceleration, bounds estimated acceleration to ± 0.1 Mbps/s^2 to limit noise amplification, and predicts six seconds ahead; it opens enough additional paths to keep forecast demand below 80% of active capacity. Active paths share traffic through the same bounded capacity allocator, with the critical class allocated first. This is a simplified policy comparison, not an implementation of MPTCP, 3GPP ATSSS, or a public-safety system.

The main outcomes are application-level multimedia QoS, not delivered goodput. In each one-second control step, the simulator produces 30 video frames per second, each represented by 14 independently transmitted segments, and 50 audio packets per second. A video frame is considered on time only if all segments arrive within a 150 ms deadline. Audio packets are duplicated only by

the predictive media-assist policy, over up to two active paths; other policies send one audio copy. A packet counts as on time if at least one eligible copy arrives within 50 ms. Path delay includes configured base latency, a utilization-dependent queue term, and random variation; packet loss is sampled independently. Video stall time is the sum of missed 1/30-second frame intervals, while a freeze event is a run of missed frames lasting at least 150 ms. Audio delay variation is the p95 absolute difference between consecutive received-copy delays. The p95 frame delay is computed over reconstructed frames, including late frames but excluding frames with lost segments. The separate handover-gap metric counts control steps where a single-path transition has no selected route; it isolates modeled route-switch interruption from congestion-induced late or lost media. These deadlines and packetization are scenario assumptions, not certified application or public-safety requirements.

Thirty-replicate mean results are: Here, V/J denotes p95 reconstructed-video-frame delay and p95 audio delay variation, respectively. Media-assist packet overhead is the number of duplicate video-segment and audio-packet transmissions divided by the number of original media packets; it is a packet-count ratio, not a byte or capacity percentage.

Policy	Video on time	Audio on time	Stall total / max (s)	p95 V/J (ms)	Handover gap
Single path	43.64%	39.79%	101.44 / 95.93	138.6 / 9.21	4.0 s
Reactive	97.13%	99.19%	5.17 / 1.00	137.1 / 2.24	1.0 s
Predictive	97.72%	99.75%	4.10 / 0.07	137.3 / 2.25	0.0 s
Predictive + media assist	99.18%	99.90%	1.47 / 0.05	48.9 / 2.30	0.0 s

The single-path policy has two modeled transitions, each with a two-second interval with no selected path: four seconds total and 120 missed video frames at 30 frames/s. Reactive multipath has a one-second standby-path activation gap (30 frames), while the predictive policy has none under this schedule. The single-path's much larger cumulative stall and 95.93-second longest freeze also reflect later capacity/queue overload in the scenario; they must not be attributed to handover alone. Multipath avoids most or all of these modeled switch gaps. Acceleration forecasting improves video on-time delivery by 0.59 percentage points and audio by 0.56 points versus reactive multipath, and reduces cumulative stall by 1.07 seconds. Its p95 video delay and audio variation are effectively unchanged, so the predictive policy alone does not show a broad QoS advantage over reactive multipath.

Adding media assist improves video on-time delivery by a further 1.46 percentage points and audio by 0.15 points versus predictive multipath, while reducing total stall by 2.63 seconds and p95 reconstructed-frame delay by 88.3 ms. The tradeoff is 47.04% duplicate packets relative to original media packets and a mean 1.862 Mbps of duplicated media payload in this scenario. The configured source media rates are 4 Mbps video and 0.128 Mbps audio; the duplication rate measures selected

packet copies, not the full source rate. Duplicate payload load contributes to modeled utilization for subsequent media packets, but the simplified allocator does not feed it back into bulk-flow allocation. These are synthetic QoS and resource-model results, not evidence that a deployed transport accelerator would achieve the same gains.

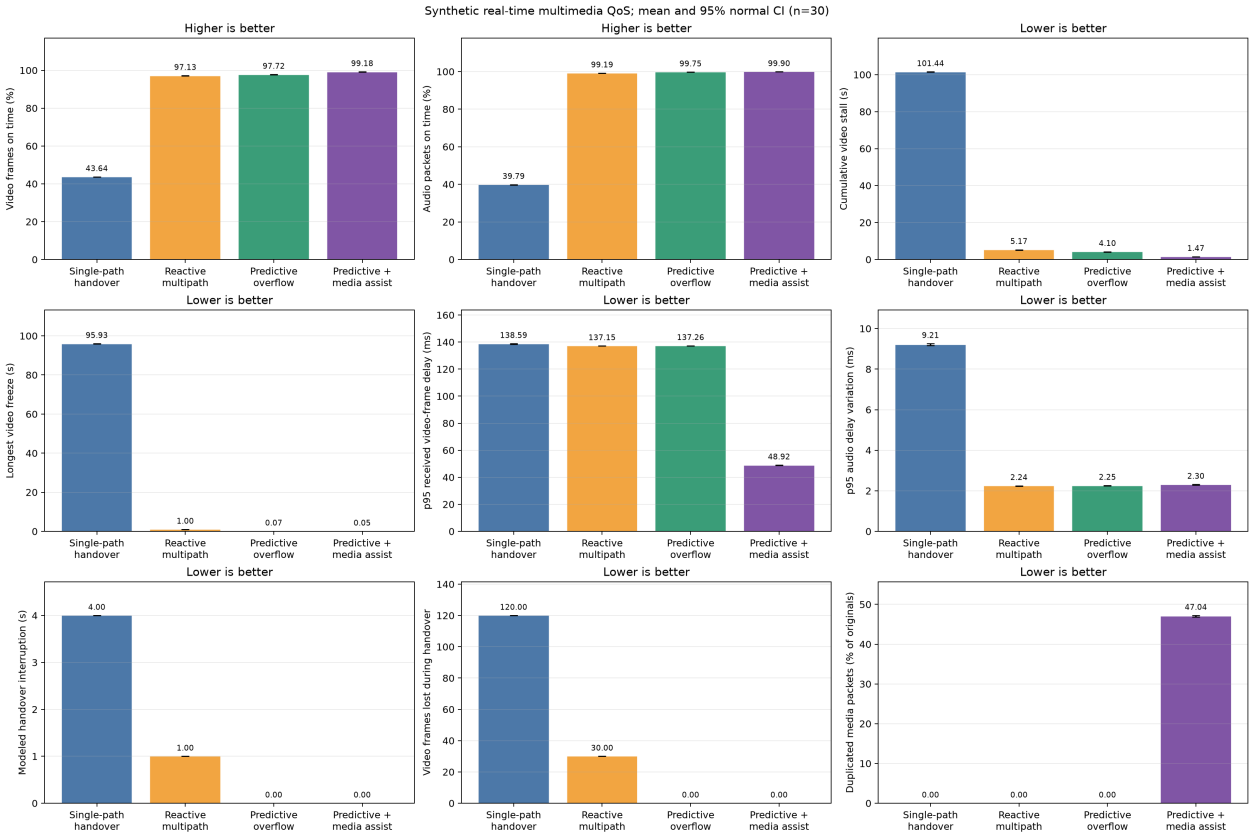


Figure 7: Real-time multimedia QoS: frame and packet delivery, stalls, delay, jitter, and handover interruption

The figure and CSV report normal-approximation 95% confidence intervals across the 30 seeds. These results illustrate the mechanism under the stated assumptions; they are not deployment measurements or an estimate of real public-safety availability. In particular, multipath activation is modeled without path setup delay, the synthetic transport omits protocol retransmission and reordering, and the Wi-Fi, mobile, HF, and satellite path characteristics are not measured. A real evaluation must measure app-level voice/video continuity, one-way latency, jitter, deadline misses, recovery time, and correlated failure using a fair 5G multi-access baseline where available, plus operator-approved 3GPP mission-critical service requirements [13]. The CSV with replicate-level summary statistics is `figures/public_safety_benchmark_summary.csv`.

10.3 Cooperative path and multicloud reorganization experiment

The prior coarse comparison mixed route forecasting, replica preparation, setup delay, and packet duplication in only two policies. Its bars therefore could not identify which change caused an

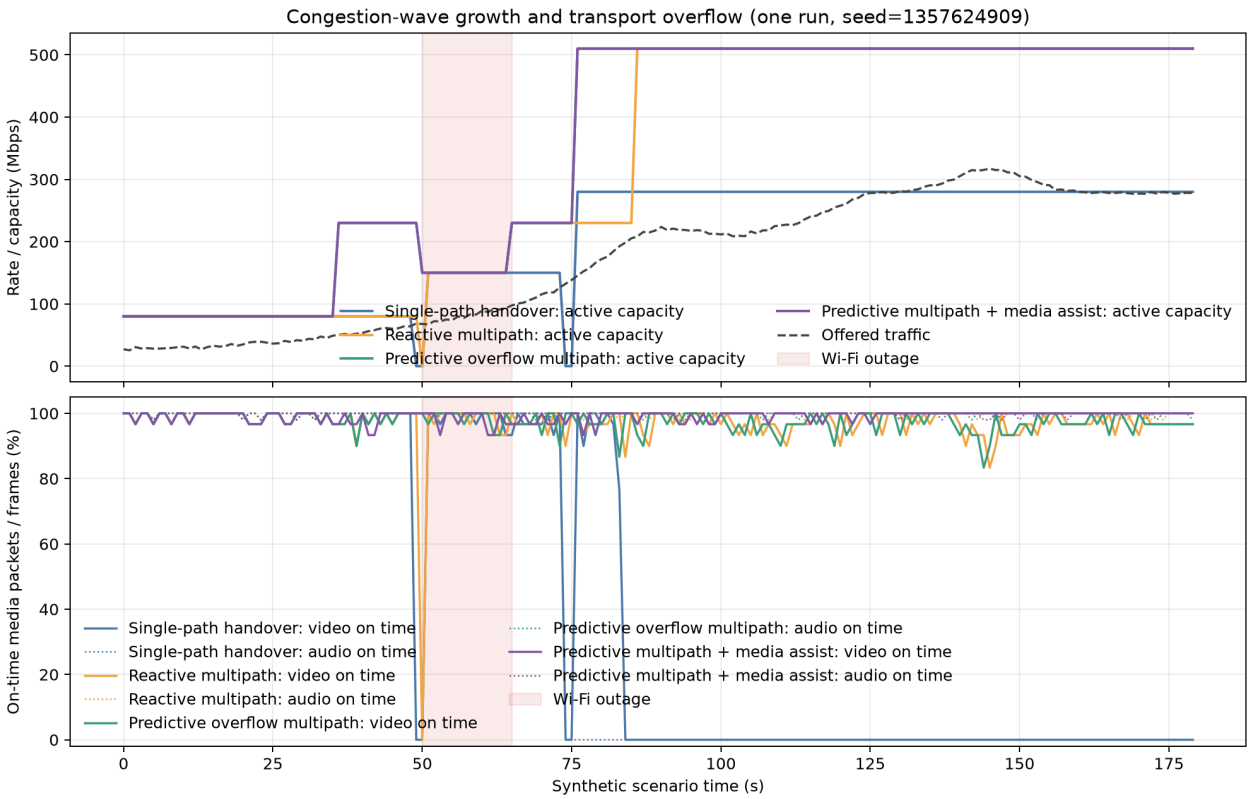


Figure 8: Congestion-wave capacity and on-time real-time media delivery over time

apparent gain, and its one-packet video-frame abstraction was not a frame-completion QoS metric. The revised experiment uses six matched policy arms and three paired scenario profiles to separate those mechanisms:

Policy arm	Predictive path activation	Early cloud preparation	Selective media duplication
Reactive baseline	No	No	No
Reactive with selective duplication	No	No	Yes
Predictive paths only	Yes	No	No
Prepared cloud only	No	Yes	No
Predictive paths and prepared cloud	Yes	Yes	No
Full cooperative	Yes	Yes	Yes

All arms use the same link/cloud capacities, event schedules, application workload, and paired demand seeds within each scenario. The scenarios are nominal; 25% higher offered load; and a correlated outage in which cellular fails with fiber, demand is 10% higher, and modeled path/cloud setup times are 50% longer. They are explicitly synthetic stress profiles, not calibrated representations of 5G or 6G systems.

This is a custom discrete-event abstraction, not a calibrated physical-layer, radio-access, or standards-complete protocol simulator. The paired arms help attribute effects to modeled policy components, but they do not establish fidelity against ns-3, OMNeT++/INET, Sionna, a protocol implementation, or field measurements. In particular, the modeled capacity, delay, loss, correlation, queue service, and cloud startup distributions are scenario inputs. No external simulator comparison or independent replication has been performed.

The transport portion is implemented with SimPy. Each link has a bounded 256-packet queue, a packet serialization rate derived from its time-varying capacity, propagation delay, jitter, and a configured base/congestion loss probability. Audio packets have priority over video segments in each link's queue. Video runs at 30 frames/s, each frame consists of four required segments, and a frame is counted successful only if every segment arrives within 150 ms; audio runs at 50 packets/s with a 50 ms deadline. Per-copy loss, frame reconstruction, interarrival variation, stalls, and recovery after the two modeled faults are measured separately. Selective duplication uses the same risk rule in the reactive-duplication and cooperative arms, and duplicate bytes enter the same path queues, so redundancy consumes modeled capacity rather than being free.

Bulk traffic is still a per-second fluid approximation. It influences link service rate and total capacity delivery but is not represented as individual packets. Cloud region A has a health/capacity schedule; region B has an abstract startup delay and capacity-time budget. A cloud service recovery measurement is based on the first one-second interval after region-A failure in which at least 95% of generated video frames meet their deadline. Runs without recovery before the horizon are marked as censored in the CSV. This does not simulate a Kubernetes API, application state replication, cloud queues, or provider-specific failover.

For each scenario and policy, the CSV reports the mean and normal-approximation 95% interval across 20 seeds, plus paired mean differences and intervals relative to the reactive baseline. Paired intervals reflect seed-to-seed variation under these fixed assumptions only. They do not include uncertainty in path capacities, delay/loss distributions, failure schedules, deadlines, controller tuning, or the choice of scenario. The outputs are the benchmark CSV, QoS comparison, resource tradeoffs, paired effect-size plot, and paired-seed time series.

Selected 20-seed mean results make the separate contributions clearer:

Policy	Video on time	Cloud recovery
Reactive baseline	90.11%	20.00 s
Full cooperative	98.46%	0.00 s
Reactive + selective duplication	90.18%	20.00 s

Policy	Video on time	Cloud recovery
Predictive paths only	90.23%	20.00 s
Prepared cloud only	98.35%	0.05 s
Predictive paths + prepared cloud, no duplication	98.46%	0.00 s

Correlated-outage policy	Video on time	p95 frame delay	Duplicate payload
Reactive baseline	87.70%	129.46 ms	0.00%
Full cooperative	98.56%	32.60 ms	5.52%
Reactive + selective duplication	88.25%	32.75 ms	7.25%
Predictive paths only	87.87%	129.47 ms	0.00%
Prepared cloud only	98.00%	128.94 ms	0.00%
Predictive paths + prepared cloud, no duplication	98.16%	128.95 ms	0.00%

Under nominal and peak demand, cloud pre-warming accounts for nearly all of the video on-time increase: the paired nominal gain is 8.24 percentage points for prepared-cloud-only versus reactive, while predictive paths alone gain 0.13 points. Adding predictive paths to prepared cloud does not improve the rounded nominal on-time result. In the correlated-outage scenario, prepared cloud alone increases video on-time from 87.70% to 98.00%, but its p95 delay remains about 129 ms. Selective duplication in the full cooperative arm cuts that conditional p95 to 32.60 ms while adding 5.52% duplicate payload; its video on-time advantage over prepared-cloud-only is only 0.56 percentage points. Reactive selective duplication also lowers conditional p95 delay, but does not repair the cloud recovery gap or materially improve on-time delivery. Thus, this model supports a specific, assumption-dependent cloud-readiness effect and a duplication/tail-delay tradeoff, not a demonstrated benefit from predictive routing by itself.

The p95 frame-delay statistic is calculated only for frames that reconstruct; it must be read alongside the on-time percentage and is not a standalone continuity measure. Duplicate payload is the duplicated media bytes as a percentage of original media payload. The plotted confidence intervals quantify replicate variability for these fixed assumptions, not uncertainty about whether those assumptions match a real network.

This is a more detailed synthetic experiment, not yet an empirically validated network simulator. Queue discipline, source traffic, path loss, cloud startup, and fault timing remain stipulated. Its results can show how the implemented rules behave under these explicit assumptions; they cannot

establish real-network effect size, prove a brain-inspired mechanism, or show superiority over a deployed or fully specified 6G system. The model also does not train or execute AI/RL, use NPU hardware, forward real packets, or run Kubernetes. A credible next validation stage would calibrate the model with measured traces, compare against standards-capable implementations, and test the same policy arms in a controlled network emulator.

10.4 Coupled congestion control: Reno, LIA and OLIA

The evaluation agenda lists fair comparison against MPTCP congestion control as open work, because the original simulator had no congestion window at all. The repository now contains a window-level coupling model (`simulation/congestion.py`) and a comparison benchmark (`tools/benchmark_congestion_control.py`) covering uncoupled Reno (RFC 5681 additive increase, multiplicative decrease), LIA (RFC 6356 eq. 1) and OLIA (the published Khalili et al. increase rule). It is a model, not a TCP implementation: there are no ACK packets, retransmission, slow start or MPTCP schedulers, the caller supplies the RTT estimate, and a loss halves only the affected subflow. Those simplifications bound what the result can mean.

The scenario is a two-path flow with 20 Mbit/s and 5 ms RTT on path A and 8 Mbit/s and 25 ms RTT on path B, 1460-byte segments, 60 simulated seconds, 8 paired seeds, one drop-tail queue of 2×BDP per path, and at most one congestion response per RTT. Means with 95 % Student-t intervals:

Algorithm	Total (Mbit/s)	Path A (Mbit/s)	Path B (Mbit/s)	Jain index	Coupled ACKs
Reno	27.9935 ± 0.0023	19.99999 ± 0.00003	7.99348 ± 0.00230	0.8446	0.0 %
LIA	27.9927 ± 0.0028	19.99996 ± 0.00004	7.99272 ± 0.00283	0.8446	67.3 %
OLIA	26.6480 ± 0.0084	19.99999 ± 0.00003	6.64797 ± 0.00843	0.7993	100.0 %

Reno and LIA both saturate both paths and are statistically indistinguishable in practice: the paired difference is -0.0008 Mbit/s (95 % CI [-0.0013, -0.0003]), 0.003 % of the 28 Mbit/s combined capacity. OLIA reaches 1.345 Mbit/s (4.8 % of capacity) less than LIA. The mechanism is visible in the model state rather than assumed: path B's mean window is 14.2 segments against a 17.1-segment bandwidth-delay product, so B's goodput falls to 6.65 Mbit/s. In the published OLIA rule, the long-RTT path needs the largest window and therefore stays in the `max_w_paths` set; the `collected` set is then usually empty, its coupling term is zero, and B grows only through the slower Kelly/Voice term while drop-tail halving keeps it below BDP. A 300-second diagnostic run held the same steady rate, so this is an equilibrium of the model, not slow convergence.

The convergence-time metric is uninformative here (every run was already inside the ±10 % band within the first two seconds), so no convergence claim is made. Nor is this a conformance

test: the Linux out-of-tree OLIA implementation uses a different inter-loss statistic, and a kernel MPTCP stack with pacing, slow start and real ACK clocks would not necessarily reproduce these numbers. The defensible statement is conditional: in this stipulated model, coupling provides no total-throughput gain over capacity saturation, and the published OLIA increase is penalised for long-RTT paths behind shallow buffers. That is a testable hypothesis for a real stack, not evidence about MPTCP. Summary, replicate and paired-difference CSVs are committed under `figures/`.

11. From multipath handover to multicloud Kubernetes

The proposed system spans two different adaptation timescales. A real-time session controller must preserve media delivery while its usable transport changes; a service-orchestration controller must place and scale caches, relays, and application workloads across edge sites and cloud regions. These controllers exchange goals and capacity signals, but must not be collapsed into one loop: a Kubernetes pod start is far too slow to repair a packet-level handover, and rapid path steering must not create unbounded cloud replicas.

11.1 Session and transport continuity

An application session has a stable service identity independent of its current access bearer. A session endpoint or trusted edge gateway measures per-path delay, jitter, loss, available capacity, MTU, reachability, and confidence in the telemetry. A path manager maintains active and standby subflows across eligible media such as Wi-Fi, cellular, private radio, HF, and satellite. Authentication, application policy, and path MTU are checked before a path is admitted. Multipath transport such as MPTCP can use several IP paths for one byte stream where both endpoints and the network support it [11]; 3GPP access-traffic steering, switching, and splitting is a relevant 5G-system capability [12]. Neither capability implies universal support for arbitrary access combinations or identical handover behavior across devices.

The predictive controller estimates demand and path-pressure velocity and acceleration. If projected load approaches a safety margin, it can establish or reserve a secondary path and shift eligible traffic before the serving path saturates. During a failure, a pre-established path may avoid the setup gap modeled for single-path switching. Traffic may be striped when reordering and application semantics permit; deadline-sensitive audio may be duplicated over independent paths; video may use selective packet duplication or application FEC when a path forecast indicates elevated deadline risk. The policy must bound redundancy, account for its capacity cost, respect data residency and priority, and retain deterministic fallback if telemetry or inference fails.

11.2 What “TCP amplifier” should mean

“TCP amplifier” is not a standardized protocol term and should not be read as creating bandwidth. TCP retransmission can add head-of-line delay: a missing byte may hold later bytes from the receiving application. That behavior can be undesirable for interactive media even when TCP eventually delivers every byte. Much real-time audio/video uses deadline-aware transports over

UDP, including RTP/WebRTC profiles, while QUIC provides both reliable streams and datagram extensions depending on the application. The correct mechanism depends on the media stack, not a generic TCP switch.

This paper therefore uses **transport accelerator** or **media assist** for a policy layer, not a device that amplifies TCP. Its functions may include:

1. classifying flows by authenticated service policy and deadlines;
2. pacing and prioritizing packets without starving control or other users;
3. steering or striping packets over established subflows;
4. selectively duplicating only packets whose predicted path risk justifies the extra bytes, or adding bounded application-layer FEC;
5. measuring late/lost frames, jitter, freezes, recovery time, and redundancy overhead, then reducing assist when path conditions improve.

For legacy TCP, a deployment could use endpoint-supported MPTCP or an explicitly configured transport proxy. A split-TCP proxy changes end-to-end transport semantics and may conflict with end-to-end encryption, congestion control, authentication, and application assumptions; it must not silently acknowledge or discard data on behalf of an endpoint. This simulator models deadline-aware media packet duplication and path selection only. It does not implement TCP, MPTCP, QUIC, WebRTC, FEC, proxying, or real packet forwarding.

In the public-safety synthetic benchmark, predictive multipath with media assist is compared with predictive multipath alone. At root seed 2026 over 30 replicates, video frames on time increase from 97.72% to 99.18%, audio packets on time from 99.75% to 99.90%, and total video stall falls from 4.10 s to 1.47 s. p95 received video-frame delay falls from 137.3 ms to 48.9 ms, while p95 audio delay variation changes from 2.25 ms to 2.30 ms. The modeled assist duplicates 47.04% as many media packets as original packets and averages 1.862 Mbps of duplicated payload in this scenario. This is a QoS/overhead tradeoff under synthetic assumptions, not a claim that a real transport accelerator achieves these values.

12. AI- and NPU-assisted TCP transport acceleration

"TCP amplification" is best treated as a proposed service name, not a literal increase in a TCP connection's capacity. A neural processing unit cannot add link bandwidth, remove propagation delay, or make lost bytes arrive sooner. The technically meaningful goal is **AI-assisted transport optimization**: use measured and predicted conditions to select a safe combination of path, pacing, scheduling, and application-level protection while preserving TCP's reliability and congestion response. TCP behavior is defined by the standardized transport protocol, and congestion control is an essential response to shared bottlenecks [16, 17]. An accelerator must not defeat those controls by injecting acknowledgements, hiding loss, or encouraging a sender to exceed a safe rate.

Functional architecture

As introduced in Section 11.2, the design separates three functions that are often conflated by the word "NPU":

1. **Packet-forwarding hardware:** a switch/router ASIC or network processing unit (NPU) classifies, queues, schedules, filters, and forwards packets at line rate according to installed policy.
2. **Neural inference hardware:** a neural NPU, GPU, or CPU evaluates a compact model over summarized telemetry. It recommends path weights, a standby activation, a pacing envelope, or selective media protection.
3. **Endpoint transport:** the TCP sender and receiver retain sequence, acknowledgement, retransmission, and congestion-control state. A network controller may configure supported endpoint or multipath behavior, but it does not silently replace endpoint semantics.

The neural model should generally operate on a control interval (for example, hundreds of milliseconds to seconds), not infer separately for every packet. Packet actions need predictable latency and remain in the router's deterministic forwarding pipeline. A model can consume windowed features such as RTT and RTT variation, ECN marks, queue occupancy and trend, loss and retransmission rates, ACK spacing, path availability, offered load, and application class. It emits bounded recommendations such as "prepare path B," "reduce this flow's pacing ceiling," or "duplicate only these deadline-bound datagrams." The policy engine checks authorization, topology, congestion, capacity, and service constraints before programming the forwarding hardware or a supported endpoint API.

Where AI may help and where it must not

Prediction may help distinguish a rising queue from a transient sample, anticipate a path outage, rank eligible paths by deadline risk and cost, or recommend a cautious pacing profile. For MPTCP-capable endpoints, supported subflows can be scheduled according to policy; RFC 8684 specifies the endpoint-level multipath TCP extensions [11]. For a conventional single-path TCP flow, changing the route may alter RTT, reordering, and failure behavior. An in-network device should not assume it can transparently split or migrate that connection while retaining semantics. An explicit proxy creates separate transport legs and therefore changes the end-to-end trust and failure model.

The controller must not treat every loss as proof of congestion or every idle queue as permission to increase the sending rate. Wireless fading, mobility, buffer overflow, route changes, and correlated failures can produce different signals; an incorrectly aggressive model can worsen congestion for other flows. Consequently:

- congestion-window evolution remains governed by a standards-compliant endpoint congestion controller (for example, CUBIC where implemented) [18];
- network pacing and prioritization respect aggregate link capacity, fairness, and configured service budgets;
- retransmission remains the responsibility of TCP endpoints; the network may provide telemetry but does not spoof ACKs or discard acknowledged data;

- packet duplication or FEC is applied only to application data whose deadline semantics support it, and its bytes are charged to the same capacity budget;
- policy is fail-safe: stale telemetry, model failure, overload, or accelerator unavailability returns control to the established non-AI forwarding and congestion-control path.

For encrypted traffic, classification must rely on authorized metadata, explicit endpoint signaling, or service policy rather than payload inspection. Model features and logs should avoid storing unnecessary user data. Signed/versioned models, provenance checks, sandboxed inference, rollback, operator override, and rate-limited actions are required before considering an autonomous deployment.

Training, placement, and Kubernetes integration

Training should use diverse measured traces spanning wired, cellular, Wi-Fi, microwave, and satellite conditions, including congestion, random loss, mobility, route changes, and correlated failures. Synthetic pretraining can initialize a model but cannot establish safe thresholds or generalization. Validation should use held-out networks, calibrated uncertainty, offline replay, and shadow mode before any recommendation is allowed to change live traffic. Online reinforcement learning must not explore by violating congestion, availability, or safety constraints; constrained contextual selection or offline policy evaluation is a safer initial research path.

In the multicloud architecture, regional Kubernetes clusters can deploy versioned model services and telemetry aggregators near routers. A global planner can distribute signed model artifacts and compatible policy bundles according to region, hardware capability, and data-residency rules. Local inference must continue when cloud connectivity is lost, and model rollout or pod autoscaling must not sit in the packet recovery loop. Kubernetes schedules the control-plane services and manages their lifecycle; router forwarding hardware performs the time-critical packet actions. Hardware accelerators must be explicitly inventoried and exposed through validated vendor drivers or device plugins. An NPU cannot be assumed merely because a cluster is Kubernetes-managed.

Evaluation criteria

An evaluation should compare a standards-compliant TCP congestion-control baseline, supported MPTCP scheduling where both endpoints permit it, static operator policy, reactive telemetry-based steering, and the proposed AI-assisted recommendation with deterministic safety guards. Replay paired traces with identical offered load and failure timing. Measure completion time, goodput, RTT distribution, retransmissions, ECN, queue delay, loss, fairness against competing flows, handover recovery, energy, inference latency, and model/action overhead. Include counterexamples: bad forecasts, model outage, route asymmetry, non-congestion loss, sudden bottleneck collapse, encrypted flows, and adversarial telemetry. Report whether QoS improves without harming other flows or violating congestion-control rules.

The current repository does not implement this TCP/NPU controller. It has a synthetic ONNX congestion forecaster and an optional inference-provider path, but no connection to TCP state, live switch queues, a forwarding ASIC, or Kubernetes model deployment. Accordingly, the concept is a

research chapter and evaluation plan, not a demonstrated TCP amplifier.

13. Multicloud and Kubernetes control plane

Each edge, regional, and cloud location operates an independently managed Kubernetes cluster. A global placement service may observe summarized capacity, cache hit/miss demand, accelerator availability, service latency, egress price, trust domain, and residency constraints through authenticated cluster agents. It selects a target region or replica set; a local cluster controller then reconciles desired state into Deployments, StatefulSets, Services, storage claims, and node-local cache workloads. Kubernetes HPA or a custom-metrics scaler can adjust replica counts from queue depth, request rate, or forecast demand, within configured min/max replicas and resource quotas. Topology-aware placement and affinity keep cooperating data-plane components close when that improves service.

The global service is not assumed to be a magical Kubernetes federation. Clusters have separate API servers, identities, failure domains, and networking. Cross-cluster service discovery, workload identity, policy, secrets, image/model provenance, service connectivity, and failover must be explicitly selected and validated. Cache placement is not the same operation as relocating a stateful service: cache objects can often be copied asynchronously after integrity and policy checks, while stateful workloads require consistency, ownership, and recovery semantics. The global planner should move content and pre-warm capacity ahead of demand where economical, but keep local serving and failover operational during cloud control-plane partition.

14. Coupling, safety, and evaluation

The end-to-end control loop is: (1) observe path and service QoS; (2) predict load and deadline risk; (3) admit an alternate authenticated path; (4) steer or selectively protect media; (5) report regional demand and resource pressure; (6) request cache replication or bounded Kubernetes scale-out; and (7) verify the resulting application QoS and cost. Path steering operates on packet-to-second timescales; orchestration and cache placement generally operate more slowly. Hysteresis, rate limits, budgets, and ownership boundaries are needed to prevent competing controllers from oscillating or scaling without limit. NPU inference may recommend actions, but authorization, rate limits, placement constraints, and fast failover remain deterministic and operator-controlled.

Evaluation should compare single-path handover, standards-capable multipath where available, reactive multipath, predictive multipath, and predictive multipath plus media assist. Replay shared Wi-Fi/mobile/HF/satellite traces with controlled outages and correlated failures. Measure session interruption duration, video frame deadlines, freezes, audio packet deadlines, jitter, tail latency, reordering, retransmissions or redundant bytes, energy, cost, and Kubernetes scale/placement delay. Include cases where duplication harms congestion, policies prohibit a route, or a cloud region is unavailable. The decision criterion is sustained application QoS within a resource and policy budget, not peak goodput in isolation.

15. Prototype status and limitations

The accompanying repository provides starting artifacts, not experimental evidence for planetary-scale operation. It is an early-stage single-author research prototype. This paper has not undergone peer review, and no independent replication is claimed. Its literature review provides technical context, not evidence of community endorsement, adoption, or novelty of the integrated design. As of this version there are no external issues, pull requests, forks, or citations; the validation, replication and citation artifacts (`docs/external-validation.md`, `REPLICATION.md`, `CITATION.cff` and the issue forms under `.github/ISSUE_TEMPLATE/`) exist so that such engagement can be recorded when it happens, not as evidence that it has happened.

Capability	Current prototype	Research gap
Multi-path mesh	In-process allocation plan using configured link attributes	Real forwarding, shared bottlenecks, transport behavior, and network testbed
Network simulator fidelity	Custom Python/SimPy event and queue abstractions with CSV KPI interchange; scenario S1 (two-path handover, CBR UDP, 100-packet FIFO) cross-validated against ns-3.49 seed by seed	Wireless PHY/MAC validation (ns-3 Wi-Fi/mmWave or Sionna), TCP/MPTCP transport, OMNeT++/INET replication, and trace calibration of the remaining parameters
UDP packet path	Actual socket datagrams over local IPv4 loopback	Multi-host/NIC transmission, physical media, loss/reorder emulation, and scale
Forecasting	Finite-difference velocity/acceleration extrapolation in the policy benchmarks; a separate small MLP pretrained on deterministic synthetic data	Compare against persistence and statistical/learned baselines using measured, temporally separated traces; quantify uncertainty and regime-shift behavior
NPU execution	Optional ONNX provider path with profile checks	Vendor hardware/runtime validation and physical-device evidence
CDN adaptation	Simulated demand forecasting, cache, and replica decisions	Production cache integration, freshness, privacy, and economic evaluation
Cooperative recovery	Deterministic paired path/cloud failure benchmark and one-factor sensitivity runner	Trained distributed AI, calibrated failure behavior, and real Kubernetes/cloud integration

Capability	Current prototype	Research gap
Orbital infrastructure	Architectural concept only	Orbital dynamics, spacecraft subsystem model, mission cost, and field data
Quantum resource	Illustrative key-buffer abstraction	Protocol-accurate quantum network and security analysis

The current model's synthetic validation result is not a measurement of real-network accuracy. Likewise, a local UDP benchmark is not a measurement of wireless, satellite, microwave, or wide-area performance. The system has not been shown to improve global coverage, affordability, latency, resilience, or environmental impact.

Implementation and software-test status

The local validation workflow is implemented in the repository. It includes measured-trace replay with explicit input checks, run-level replicate exports, paired one-factor-at-a-time sensitivity sweeps, and CSV adapters for normalizing and comparing simulator or testbed results by scenario, policy, metric, and seed. The comparison tool requires at least two matched seed pairs to report a paired uncertainty interval and rejects unit mismatches. Automated tests cover trace validation and replay, parameter scaling, sensitivity outputs, external-result normalization, paired comparisons, and rejection of undersized comparisons.

On 2 November 2026, the full local Python test suite completed with **93 tests passing** (plus 7 parameterised subtests; `py -m pytest`), including the validation-workflow, link-queue, controlled-benchmark and coupled congestion-control tests. A measured-trace smoke run using generated temporary input, two replicates, and one custom scenario completed and produced the expected outputs for all six policies. These results establish software behaviour for the tested code paths only. They are not wireless PHY validation, operational-trace calibration, MPTCP conformance, or testbed evidence, and they do not validate physical-network fidelity.

The policy benchmark uses a simple second-order extrapolation, $\hat{x}(t+H) = x(t) + vH + \frac{1}{2}aH^2$, with finite-difference estimates of velocity and acceleration. This is an interpretable baseline, not a learned forecasting method and not evidence that traffic follows constant acceleration. The synthetic ONNX MLP is a separate prototype artifact; it is trained on synthetic telemetry and is not the predictor used to establish the paired benchmark results. Neither model has been validated on representative operational traces.

The current benchmark set also remains narrower than the proposed system. Its path allocator and failure schedules are abstractions; its cooperative failure study tests deterministic path activation, cloud pre-warming, and selective duplication, not trained distributed-AI reorganization or live multicloud Kubernetes. The earlier vision benchmark shared an allocator between reactive and predictive policies and gave the predictive arm larger cache gains, so similar delivered goodput was expected by design. The controlled re-evaluation in §10.1 fixes both issues and changes the

conclusion rather than rescuing it: with independent allocator instances and cache gains held constant, pressure prediction reduces delivered goodput in every scenario while cache prediction supplies the measured gains. These experiments support only conditional observations about their implemented mechanisms, not claims about a complete 6G/7G system, and the predictive-goodput claim is withdrawn.

The next work should first cross-check one wireless link/handover scenario in a network-level simulator with a real PHY/MAC model and calibrate inputs against documented measurements. It should then extend the transport comparison to protocol-compatible MPTCP baselines, including LIA/OLIA congestion control where appropriate and established schedulers, with matched workloads and resource accounting; the repository now includes a coupled-window LIA/OLIA model and a comparison benchmark, but those results are model-level and not yet cross-checked against a kernel MPTCP implementation. The implemented one-factor sensitivity runner varies demand, capacity, delay, loss, setup/startup scaling, and forecast horizon; broader factorial or global analysis over correlated failures, forecast error, and cloud assumptions remains necessary. External review, independent replication, and controlled emulation/testbed evaluation remain future work, not completed project milestones.

16. Surgical, rate-controlled state updates

The observation layer in §5 specifies normalised path pressure and its first two derivatives, and names acceleration-triggered cross-medium overflow as the regime of interest. It does not specify the **control law** that maps those observations to changes in network state. This section supplies that law and tests it; §16.4 reports where measurement contradicts the design intent.

16.1 State plane

A node's editable state is the vector `S = (alpha, queue policy, FIB overrides)`, where `alpha in [0,1]` splits capacity between forwarding (`(1 - alpha) C_n`) and serving (`alpha C_n`), the queue policy sets per-class weighted-fair-queueing weights and byte limits, and FIB overrides hold per-flow next hops layered over a loop-free name table. Every mutation passes through a single ledger, so the speed and acceleration of state change are measured at the point of edit rather than reconstructed afterwards. This matters: the dependent variable of H2 is precisely that trajectory.

16.2 Kinematic monitor and the spike trigger

Each node keeps exponentially weighted baselines of pressure, pressure velocity, and acceleration, and derives its threshold from the exponentially weighted magnitude of its own acceleration residual, with multiplier `k = 4` and a physical floor. The intent is that a steady network is silent without a hand-tuned constant: the threshold tracks the measured noise floor. An alpha-beta filter variant is evaluated alongside it, because the argument that a trend-following filter beats a moving average has been made for RTT prediction.

16.3 Control law

Two distinct forms of rate control are applied, because they bound different things. *Discrete edits* – promoting one `(content, source-region)` class into its own weighted lane at the routers it crosses – are applied in a single step and bounded by a **per-node token bucket** with a sustained rate and a burst allowance; a node that has just fired cannot fire again for `|dS| / v_max` seconds. *Continuous knobs* – `alpha` and buffer scale – move through a **double integrator** with explicit speed and acceleration caps.

Hysteresis is a **pressure** test, not an acceleration test. A sustained flood has zero acceleration and high pressure, so releasing on calm acceleration un-isolates in the middle of the attack. Release additionally requires the culprit class's arrival rate to have stopped for several control ticks.

16.4 Measurement

The testbed is a 100-node three-medium fabric (SimPy, not ns-3): 1 Gbps / 2 ms fiber, 100 Mbps / 10 ms / 1 % microwave, and a 10 Mbps / 500 ms / MTU 576 satellite channel. Four origins are single-uplink leaves behind 100 Mbps microwave last miles, so a name-targeted flood necessarily strikes a bottleneck that legitimate traffic also uses. Eleven arms are compared over three experiments and 30 paired seeds (840 runs): two baselines, four ablations, one confound control, two composability arms, and a hyper-specialisation arm.

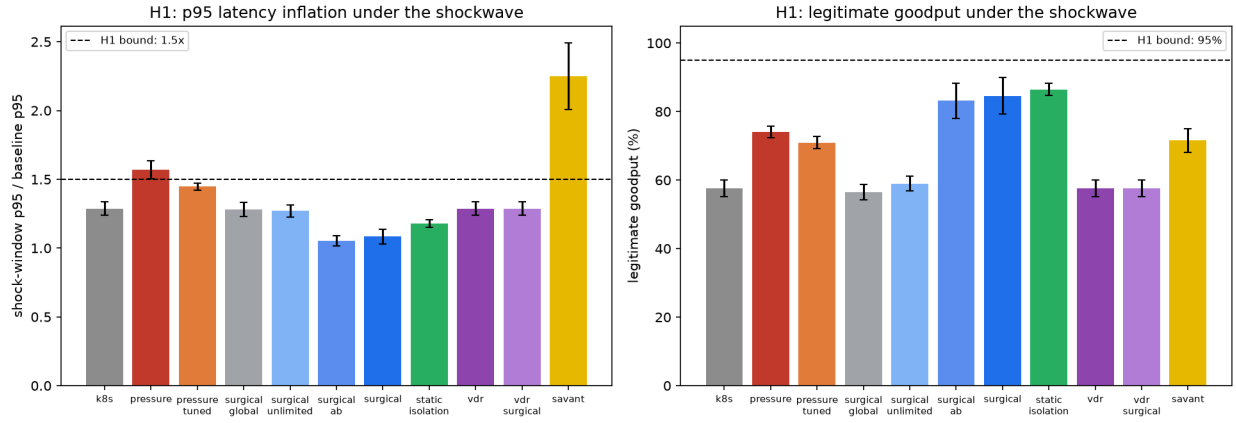


Figure 9: Surgical-state-update headline

H1 is refuted on two of four clauses. The surgical arm holds a 1.08 p95 latency ratio where the global pressure controller degrades to 1.57, and recovers 11 percentage points of legitimate goodput (84.6 % versus 74.1 %). But legitimate goodput does not exceed 95 %, and the global controller does not starve the fabric to 2.2x.

The confound control decides the interpretation. Per-flow fair queueing installed for every flow, with no sensing and no trigger at all, reaches 86.5 % goodput and a 1.18 ratio. Almost all of the quality-of-service recovery is therefore attributable to queue isolation – a deployment practice predating this work – not to the kinematic trigger. What the trigger buys is **state economy**: 45

edits and 12 isolated classes against 215 edits and 215 permanently installed classes. A causal claim that acceleration sensing produces the resilience would be wrong on this evidence.

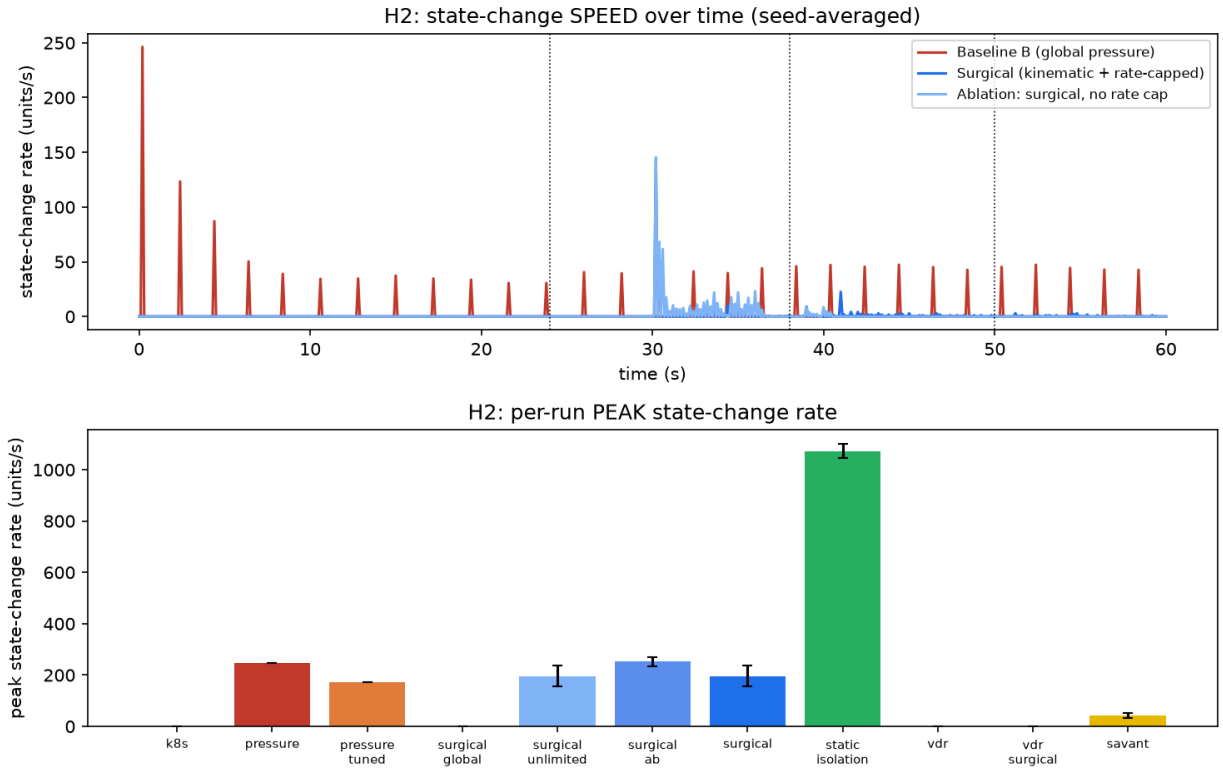


Figure 10: Surgical-state-change

H2 is supported in part. The sustained state-change rate falls to 21 % of the global controller's with 68x fewer edits, and responsiveness improves rather than degrades. The seismograph prediction is confirmed for sustained activity: a flat line interrupted by one short spike per edit. It is refuted for peak magnitude, 0.80x rather than below half. The reason is structural: a coordinated mitigation edits roughly twelve routers inside one control tick, so the fleet-wide burst is large even though each node's budget is small. A per-node limiter bounds how often a node changes state; it cannot bound a synchronous burst across nodes without a global limiter – which would reintroduce the centralised controller the design avoids.

H3 is refuted, and the mechanism is instructive. At a fiber-to-satellite boundary the reassembly window saturates in every arm: 70.3 % of the run above 80 % occupancy for the surgical arm against 70.2 % for the static baseline, a difference well inside the confidence interval. Two findings follow. First, the reorder window is consumed by the number of in-flight fragmented objects over a deep satellite buffer, and a queue-class edit at the ingress does not shorten that buffer; legitimate objects alone can fill it. Second, the acceleration trigger is *masked* in this regime: the two-second satellite buffer produces a periodic pressure sawtooth (0.04 to 0.81), inflating the adaptive threshold to roughly 8 units/sˆ2 while the flood's residual is only about 3.1, so the detector goes blind exactly where the vision paper expects it to act. An adaptive noise floor derived from

residual magnitude adapts to the pathology it is meant to detect.

H4 (compensatory hyper-specialisation) is supported as stated, with a measured threshold. It is swept over its gain and reported in section 17; in brief, asymmetric hyper-allocation beats symmetric load balancing and smooth gradient actuation on affected-name delivery, peaks at `alpha -> 0.85`, and collapses into cascading starvation beyond it.

The spatial/temporal composability pairing is untested. With single-uplink origins there is no admissible loop-free alternative to the origin uplink, so a pressure-weighted potential gradient never differs from the static forwarding table and both composability arms are numerically identical to the static baseline. A temporal control layer cannot rate-limit a target that does not exist; the pairing must be re-run on a topology with genuine path diversity.

16.5 What this changes in the thesis

The measurement sharpens the contribution. Kinematic sensing is not the novelty: the previous iteration's controller already consumed velocity and acceleration, and its failure is one of *actuation*, not observation. The defensible claim is that **where** state is edited and **how often** it may move are what separate a resilient fabric from a reactive one – localised, rate-capped edits achieve most of the available quality-of-service recovery at an order of magnitude less control-plane churn, while global proportional scaling on the same signal destroys it. Presented as a claim about the trigger's causal power, the paradigm is refuted by its own control arm.

Full detail, including five degenerate calibration configurations that were found and corrected, is in `surgical-rate-controlled.md` and `surgical-results.md`.

17. Compensatory hyper-specialisation (the "savant" mechanism)

Standard self-scaling networks (Kubernetes, MEC) employ symmetric, proportional scaling in response to load. This section proposes and tests an alternative inspired by biological neuroplasticity: **compensatory hyper-specialisation**. When a node detects a kinematic shockwave – the "injury" – it may intentionally shed its general routing role and execute a severe, asymmetric shift in its role fraction `alpha`, dedicating its local resources to the affected circuit. The biological framing is an analogy, not evidence; translated into control terms the mechanism is an asymmetric, extreme-compensation control law with a measurable cost. The hypothesis is that it provides superior resilience for the affected flow under attack, at the measurable cost of system-wide transit efficiency, and that the trade-off has a threshold.

The mapping is deliberate and mechanical:

biological concept	network equivalent
neural tissue	the multi-medium backhaul fabric
injury or deficit	DDoS shockwave, link failure, congestion spike
neuroplastic rewiring	role fluidity, `alpha` in `[0, 1]`

biological concept	network equivalent
hyper-allocated circuit	one isolated `(name, source-region)` queue
savant skill	a node drops general transit to serve or scrub one circuit
cost of the gift	neighbours absorb shed transit; system-wide starvation

The controller isolates the culprit class with full weight, drives `alpha` toward a target through the same speed- and acceleration-capped governor described in section 16, and holds until pressure recovers. The target is swept, because a single value cannot locate a threshold.

17.1 Measurement

Five gains are compared against symmetric load balancing (Kubernetes-style static roles), the global pressure controller, a smooth gradient flux field (VDR-style), and the rate-controlled surgical arm, which is the *control*: it attempts to balance general traffic and the spike rather than trading one for the other. Thirty paired seeds; "affected" is the attacked name, "other" is every other name, and "spillover" is the throughput change at the neighbours of every `alpha`-edited node.

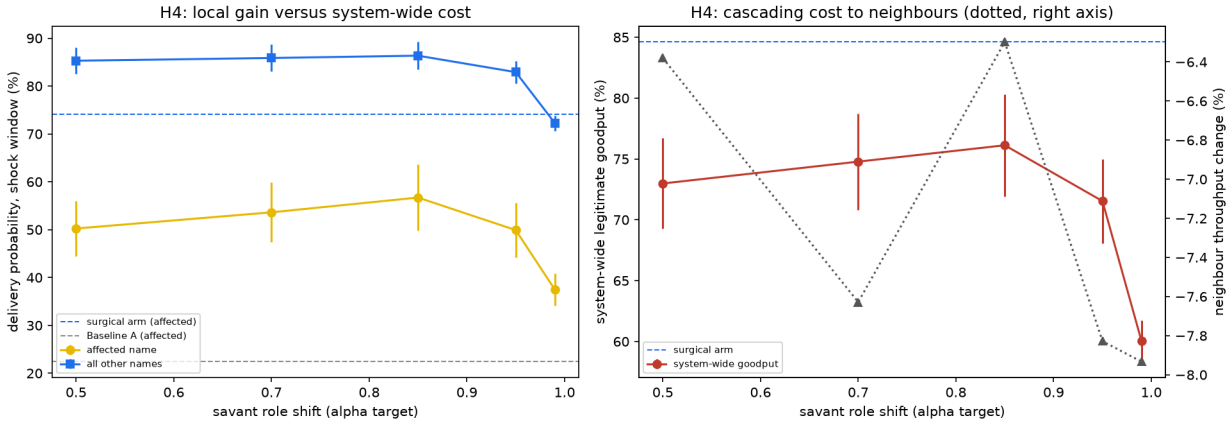


Figure 11: Savant gain sweep

arm	affected	other	system-wide	p95 ratio	spillover
Baseline A (Kubernetes-style)	22.4 %	75.5 %	57.6 %	1.29	-
Baseline B (global pressure)	55.6 %	83.4 %	74.1 %	1.57	-

arm	affected	other	system-wide	p95 ratio	spillover
VDR (smooth gradient)	22.4 %	75.5 %	57.6 %	1.29	-
Savant, `alpha -> 0.50`	50.2 %	85.3 %	73.0 %	1.18	-6.4 %
Savant, `alpha -> 0.70`	53.6 %	85.9 %	74.8 %	1.19	-7.6 %
Savant, `alpha -> 0.85`	**56.7 %**	86.3 %	**76.1 %**	1.29	-6.3 %
Savant, `alpha -> 0.95`	49.9 %	82.9 %	71.5 %	2.25	-7.8 %
Savant, `alpha -> 0.99`	37.5 %	72.3 %	60.0 %	**4.09**	-7.9 %
Rate-controlled control	**74.1 %**	**90.1 %**	**84.6 %**	**1.08**	-

17.2 Findings

The hypothesis holds against its stated comparators. Savant rewiring delivers 56.7 % of the affected name against 22.4 % for both symmetric load balancing and smooth gradient actuation, and it does so while keeping system-wide delivery above Baseline A (76.1 % against 57.6 %). Extreme, asymmetric compensation is not without value relative to symmetric scaling.

The threshold is real and quantified. Affected-name delivery is an inverted U in the gain: it rises from 50.2 % at `alpha -> 0.50` to a maximum of 56.7 % at `alpha -> 0.85`, then collapses to 49.9 % at 0.95 and 37.5 % at 0.99, where the p95 ratio reaches 4.09 and system-wide throughput falls below Baseline A. The crossing into cascading starvation lies between 0.85 and 0.95. A mechanism described as "violent" is therefore only useful while it is not too violent.

The predicted cost is larger than predicted, and it is shared. Neighbour spillover is negative at every gain, between -6.3 % and -7.9 %: when a node sheds transit its neighbours do not absorb that load at higher throughput, they deliver less than in the steady window. The cascade is not relocated, it is amplified. The biological caveat – that savants struggle with ordinary tasks – maps onto a measurable systemic penalty that appears even at the optimum gain.

But the mechanism loses to smooth actuation. The rate-controlled surgical arm delivers

74.1 % of the affected name, 90.1 % of everything else and 84.6 % system-wide at a 1.08 p95 ratio: better than the savant on all four axes at once, using a fraction of the state churn. The reason is structural rather than incidental. On this fabric the congested node is the last router before a single-uplink origin, so shedding its forwarding capacity removes capacity from the very path the affected traffic needs. Compensatory hyper-specialisation can pay only where the specialised resource sits *downstream* of the constraint – a dedicated scrubber in front of the origin – not where it consumes the transit capacity that the affected flow depends on.

The paradigm is therefore not wrong so much as locally mis-scoped: the mechanism does what it claims against symmetric scaling, has a real and locatable operating limit, and is dominated by a cheaper smooth alternative in this topology.

18. OEF compensation and multidimensional baselining

Two extensions to the surgical controller are tested as pre-registered hypotheses in `oef-mkb.md`; both are framed as control theory, with the biological language kept strictly as analogy.

18.1 The Fick-principle mapping (OEF)

The brain holds metabolic rate `CMRO2` constant when cerebral blood flow `CBF` falls by raising the oxygen extraction fraction `OEF`: `CMRO2 = CBF x OEF x Ca`. Mapped to the fabric, CBF is delivered throughput, Ca the nominal capacity, CMRO2 legitimate goodput, and OEF the weighted-fair-queueing weight of the legitimate class. When a node's windowed CBF falls below 0.8 of its moving average for three consecutive ticks, an OEF actuator raises the legitimate class's weight `1 -> oef_max`, proportional to the measured CBF deficit and bounded by a double integrator with explicit speed and acceleration caps.

The hypothesis is refuted, and the zero-sum trap is confirmed. Over 30 paired seeds the OEF arm is statistically indistinguishable from the surgical arm it extends: legitimate goodput is 0.949 with or without OEF, at a 0.92 p95 ratio. The reason is structural rather than incidental. On this fabric the congested node is the last router before a single-uplink origin and is a pure forwarder (`alpha = 0`); once the attack is name-isolated, the legitimate class already receives roughly 95 % of the link, so "extraction efficiency" has no further degree of freedom to exercise. Unlike a capillary bed, a drop-tail weighted-fair-queueing router cannot recruit additional service for one class without removing it from another, and the other class is already throttled. The OEF actuator is not inert — it reaches its full excursion and its speed limit is hit — but the excursion purchases nothing. This is the "zero-sum" boundary the concept was designed to find, and it is reported as a refutation, not hidden.

18.2 Multidimensional kinematic baselining (MKB)

The network state is a vector `S(t) = [utilisation, queue delay, loss, OEF]`. Velocity and acceleration are the curvature of a short-window manifold of `S`, and a surgical edit fires when the

acceleration vector leaves its moving-average baseline or the velocity vector turns away from its own trajectory. The intent is to detect cross-correlated failure geometry that scalar thresholds miss.

The straightforward implementation over-triggers: it issues an order of magnitude more edits than the scalar trigger, most of them before the shockwave, and legitimate goodput degrades relative to the scalar controller. The cause is instructive and is itself the finding: the state vector includes the OEF actuator's own output, so the multidimensional trigger detects the controller's own feedback as a spike — a self-referential loop that scalar thresholding sidesteps by construction. A multidimensional baseline is only as good as the choice of dimensions it watches; including an actuator's output in the baselined vector couples the observer to its own action.

Neither extension is claimed as an improvement. Both are recorded because the refutations bound the paradigm: extraction scaling is a real degree of freedom only where a node meaningfully serves traffic locally, and multidimensional triggering requires excluding self-inflicted state before it can outperform a scalar baseline.

19. Conclusion

Cooperative multi-medium backhaul offers a way to connect content placement, traffic engineering, and service continuity across terrestrial, aerial, and potentially orbital resources. The proposed temporal topology model gives the controller a shared representation of cross-layer services and constraints while leaving packet encapsulation and protocol interoperability to established standards and explicit adapters. The central proposal is a bounded, active-active control loop that uses telemetry trends to shift traffic and cache capacity before congestion becomes severe, while optimizing against explicit service, cost, energy, and policy constraints. Orbital AI edge nodes may be valuable for specific workloads, but their power, thermal, contact, cost, and sustainability limits must be treated as architecture inputs rather than future details.

This paper frames the idea as a research program, not a claim of a finished "7G" system. The path forward is measurement, and the first steps are now taken rather than promised: the transport model has been cross-checked against ns-3 for a bounded handover scenario, the queue defect that check exposed has been fixed, and the predictive goodput claim has been tested with a controlled ablation and withdrawn. What remains is harder and more valuable: validate the model against wireless PHY behaviour and measured traces, test multipath transport against protocol-compatible MPTCP baselines, quantify content-placement tradeoffs, establish safe control boundaries, and evaluate orbital scenarios against terrestrial alternatives. Progress on those questions, not terminology alone, will determine whether this vision can contribute to more resilient and widely accessible communication.

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